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Toward reliable false discovery rate control in classification problems under distribution shift

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Keywords:	Machine Learning, Error control,
Abstract:	Background: Domain shifts or batch effects can significantly degrade the performance of machine learning-based classifiers in software production without any warning. This can particularly be problematic in medical applications which involve human lives. The shift changes the shape of the prediction score distribution of the test samples compared to that of the training samples; hence, decision boundary re-calibration would be required to conform to the expected performance standards. Method and Findings: We present a simple, yet robust heuristic method to calculate valid p-values (uniformly distributed) for raw prediction scores of test samples under data shift in general classification problems. Our method is based on an adjustment of the test prediction scores so that the test null distribution (approximated distribution of the prediction scores of negative test samples) aligns to the training null distribution (actual distribution of prediction scores of the negative training samples). We termed it the Test Null Adjustment (TNA) method. Then, empirical p-values of the adjusted test prediction scores can be calculated with respect to the training null distribution to be used in any statistical applications which require test samples accompanied by valid p-values. We demonstrate the usability of our TNA method in FDR control with Benjamini-Hochberg (BH) protocol. Conclusion: The main advantage of the TNA method is that its application does not require any domain specific knowledge. It works in binary and multi-label classification scenarios, it is fully data-driven, and it operates in the 1-dimensional space of the prediction scores ensuring stable numerical calculations. We have performed four experimental tests on the biomedical image analysis domain to demonstrate that the TNA method is able to mitigate the bias in FDR induced by data shift in real-life applications. TNA is available at https://github.com/Aborevsky01/TNA
Order of Authors:	Attila Kertesz-Farkas
	Andrey Borevskiy
Opposed Reviewers:	
Response to Reviewers:	Academic Editor – Journal Requirements "1. If the reviewer comments include a recommendation to cite specific previously published works, please re- view and evaluate these publications to determine whether they are relevant and should be cited. There is no requirement to cite these works unless the editor has indicated otherwise." Neither Reviewer 2 nor Reviewer 3 recommended specific previously published works to cite, so no changes were necessary on this point. Reviewer #2 "The authors have addressed all the concerns I raised regarding the previous version.

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Therefore, I recommend
accepting this manuscript as it is."
We thank the reviewer for the careful re-evaluation of our manuscript and for the positive recommendation.

Reviewer #3
We thank the reviewer for the careful and critical reading of our manuscript, and for raising these methodological
and statistical concerns. We have split the reviewer's comments into three points and address each of them
individually below.

"1. What is the statistical rationale for assuming that the distributional shift only affects the null component
and not the signal component (positive scores)? If a batch effect or a shift in scale occurs, it typically affects the
entire distribution. The authors must provide a theoretical proof or a robust simulation-based argument showing
why this asymmetric approach does not introduce bias into the detection of true positives."

We thank the reviewer for this comment. This comment definitaly leads to an interesting discussion; it touches
on a few related aspects of TNA, so we address it below from several angles.
In our response preparation, we looked at Step 5 of the TNA algorithm and found that the previous revision
contained an implementation inconsistency there: the corrective affine adjustment was applied only to test scores
with a negative sign ($ts < 0$), while non-negative scores were left unadjusted. This was not present in our original
submission; it was introduced during the previous revision as a result of a miscommunication within our team,
and it is an error. Restricting the adjustment to one side of zero can reorder test instances relative to one
another (an adjusted negative-sign score can end up larger than an untouched positive-sign score), which then
also distorts the ground-truth reference computed from the true labels against this reordered set of scores. We
are grateful to the reviewer for a comment that led us to find and fix this.
We have corrected (reverted) Step 5 so that the affine adjustment is applied uniformly to every test score as
in our original submission, regardless of sign: $\hat{}ts = (ts - \mu\ \hat{}T0)/\sigma\ \hat{}T0 \cdot \sigma X0 + \mu X0$, for all $ts \in Ts$. Since this is now
a single, monotonically increasing affine map applied to the entire test score set, it preserves the relative order
of all test instances – no reordering artifact can occur – and it recalibrates the whole test score distribution, not
just its sub-threshold part, toward the location and scale of the training null. Therefore, TNA does not treat the
shift as confined to the null component; the corrected Step 5 adjusts the entire score distribution using the same
transform.
We re-ran all experiments reported in the manuscript with the corrected implementation. The qualitative
conclusions are unchanged: the corrected figures are hardly indistinguishable from the previous ones, and all com-
parisons between methods reported in the manuscript remain valid. We have updated the manuscript accordingly
(see Step 5 in the "TNA protocol" section).
A few of the broader statistical points raised by this comment remain relevant independent of this fix, which
we address below.
(A) On statistical power under data shift: the theoretical guarantee of the Benjamini–Hochberg procedure

	(and of the underlying p-value machinery in general? ?) depends only on the p-values of the truly null instances being (super-)uniformly distributed; no analogous requirement is placed on the p-values of the truly non-null (positive, signal, alternative hypothesis) instances. However, we note that we do not have any guarantees on the statistical power. Depending on the shift, it can result in either decreased power (if the test non-null scores move closer to the training null X0, yielding larger p-values for the true positives) or increased power (if the test non-null scores move further away from X0, yielding smaller p-values for the true positives). In the manuscript, in the TNA protocol, and in our experimental validation, we do not claim that the prediction accuracy remains the same (as in the training dataset); and we do not claim that we can fix wrong predictions for samples which were misclassified due to data shift. Our aim is to make the FDR control at the desired level less biased under data shift. (B) On the distributional assumptions of TNA: we would like to point the reviewer to the “TNA protocol” section of the manuscript, which already states precisely what TNA- and TNA+ assume, we quote it below: “The first variation, called TNA- (TNA minus), relies on the assumption that the shapes of the training null (X0) and test null (T0) distributions are ‘similar’, but there is no assumption on the form of the non-null distributions. The second variation, called TNA+ (TNA plus), relies on that the shapes of the training non-null (X1) and test non-null (T1) distributions are ‘similar’, but there is no assumption on the analytical form of the null distributions. By ‘similar’ distributions we mean that they are close in a statistical sense, such as having a small Wasserstein distance or Jensen-Shannon divergence; intuitively, their histograms can be aligned without major reshaping. TNA’s performance is robust to minor distributional shifts in practice, but it degrades gracefully as this distance increases. The proportion of the nulls may freely vary between the training and test data in the case of both TNA methods, there is no assumption on this.” That is: TNA places its similarity assumption on only one component (null for TNA-, non-null for TNA+), leaving the other component’s shift unconstrained, and the manuscript states that TNA degrades as the similarity decreases between training and test. The user can pick TNA- or TNA+ depending on whether the null or non-null component is expected to be more stable (respectively). (C) We note that, we already have simulation-based evidence, in the manuscript, of TNA behaving well under a shift that plausibly affects the entire score distribution rather than the null alone: in Example 2 (MNIST with a 10% reduction of pixel intensity applied to every test image), the perturbation acts on all test images regardless of their true label, so both the null and non-null test scores are affected; TNA nonetheless recovers accurate FDR control (Fig 3(i)). (D) On theoretical guarantees: we already discussed this with another reviewer in the first round; we sum- marize that response below for completeness (see our previous Response to Reviewers, Reviewer 2, point 1, for the full discussion): TNA is presented as a practical heuristic, not as a method with proven error bounds.

	We state in the manuscript that our method does not provide any theoretical guarantee that the FDR control with TNA EPVs becomes accurate, or conservatively or liberally biased, without additional assumptions; hence TNA is heuristic in its general form. To our knowledge, no existing method for FDR control under unconstrained, general distribution shift currently offers such guarantees either; we therefore frame TNA as a practically useful, model-agnostic alternative rather than as a theoretically certified procedure. More fundamentally, most FDR-controlling procedures (e.g., BH) only carry their theoret- ical guarantee under the assumption that the input p-values are valid, i.e., uniformly distributed under the null; in practice, this validity typically relies on an assumed parametric null (most commonly a standard normal distribution) that rarely holds exactly for real data, so FDR control breaks down not because the procedure itself is flawed, but because its premise is violated. TNA targets exactly this gap by estimating the null empirically from the training data instead of assuming a parametric form; the price of dropping that assumption is that we can only offer a heuristic argument rather than a formal guarantee, a trade-off already reflected in the manuscript's title, "toward reliable FDR control" rather than "guaranteed FDR control". If the reviewer referred to something else then we kindly ask him/her to elaborate or point to the relevant paragraph/sentence of the manuscript. "2. Conceptual Ambiguity Regarding FDR Control: The manuscript exhibits a fundamental confusion between FDR control in the context of 'Feature Selection' (multiple testing across features) and 'Sample-level Inference' (calibrating significance for a single test). The authors fail to clarify whether the primary objective of TNA is to improve feature selection stability or to ensure the validity of p-values for individual samples. This conflation undermines the statistical interpretability of the results. The authors must explicitly state the scope of their FDR control and demonstrate how the TNA method accounts for the multiplicity of tests if applied to feature selection, or conversely, justify why a sample-level adjustment is sufficient for high-dimensional feature discovery." We thank the reviewer for asking us to clarify this, and we would like to clarify: we do not perform, nor do we make any claim about, feature selection anywhere in this manuscript. TNA and the FDR control built on top of it operate entirely at the level of test instances (samples), not at the level of features. We agree that the manuscript should state this scope explicitly rather than leave it to be inferred. We add the clarifying sentences below to the Introduction. We state this in terms of the general set-up (test instances and their scores), without yet referring to our method by name, since it has not been introduced at that point in the manuscript. We emphasize that the multiplicity addressed throughout this paper is across the m test instances t1, . . . , tm, each associated with its own null hypothesis ("instance ti belongs to the negative class"); it is not a feature-selection problem, in which the multiplicity would instead be across a

	set of candidate features or variables. Each score ti is a single, already-computed scalar value produced by a machine learning classifier for one test instance, not a property of an input feature, and the classifier's input features play no further role once this score is obtained. We would like to note one further nuance, which we think is worth making explicit rather than leaving implicit. In two of our datasets, TissueNet and BCSS, the classification task is performed at the pixel level: each pixel of an image is treated as one test instance, and the model predicts a class for every pixel individually. Since the pixels are themselves the raw input to the image, the multiple testing in these two datasets is, in a sense, simultaneously over test instances ("samples", in machine-learning terminology) and over input features (pixels, in a feature-selection-style terminology); the same underlying set of entities can be described either way. However, pixel-wise image classification and semantic segmentation are conventionally discussed in the computer vision and machine-learning literature by treating each pixel as an individual classification instance (a sample), rather than by using feature-selection terminology, in which pixels would instead be framed as candidate features to be individually tested and selected. To avoid needlessly complicating the exposition, and to remain consistent with how this task is conventionally described in that literature, we discuss pixel classification throughout the manuscript as a machine-learning (sample-level) problem rather than as a feature-selection problem; we did not intend to imply anything about feature-selection multiplicity by this choice of terminology. "3. While the manuscript introduces TNA to manage FDR under distributional mismatch, the results currently lack sufficient statistical rigor. To address this, the authors should employ bootstrapping procedures on both the training and test sets to generate and report confidence intervals for all performance metrics." We thank the reviewer for this suggestion and have carried out the requested bootstrap analysis. In each of 100 replicates, we jointly resampled, with replacement, both the training data (from which X0 and X1, and hence the TNA correction, are estimated) and the test data, and recomputed the entire pipeline – the EPVs, the TNA correction, and the BH/LTT-based curves – on each resampled pair. We report the resulting variability as a shaded band (bootstrap percentile interval) around each of the previously reported curves in Figs 5 and 9. The outcome for the datasets with comparatively small test sets (PCam, CheXpert), the bootstrap bands are narrow but visible, reflecting genuine finite-sample uncertainty. For the datasets with much larger test sets (TissueNet, BCSS), the bands are so narrow that they are not visually distinguishable from the point-estimate curves themselves, since the variability of the underlying histogram-based estimates shrinks as the number of test (and training) instances grows. This confirms that the curves reported throughout the manuscript are not artifacts of a particular train/test split, and that their variability, where present, decreases exactly as expected with dataset size. We included the following text to the beginning of the results section: We qualified the stability of the methods with bootstrap analysis. For each dataset, we

	resampled both the training and the test sets 100 times with replacement, and recomputed the standard EPV, the TNA, and the LTT methods on each resampled set. The resulting variation is indicated on the plots as a band width around each curve. Datasets with a large number of samples, such as TissueNet, and BCSS, show a very small, hardly visible bandwidth on the curves below, reflecting the low sampling variability that comes with a large sample size. The PCam and CheXpert datasets have fewer samples, and consequently shows a visibly wider bandwidth resulting from this variation.
Additional Information:	
Question	**Response**
Competing Interests Use the instructions below to enter a competing interest statement for this submission. On behalf of all authors, disclose any competing interests that could be perceived to bias this work—acknowledging all financial support and any other relevant financial or non-financial competing interests. This statement is required for submission and **will appear in the published article** if the submission is accepted. Please make sure it is accurate and that any funding sources listed in your Funding Information later in the submission form are also declared in your Financial Disclosure statement. View published research articles from *PLOS ONE* for specific examples. **NO authors have competing interests** Enter: *The authors have declared that no competing interests exist.* **Authors with competing interests** Enter competing interest details beginning with this statement: *I have read the journal's policy and the authors of this manuscript have the following competing interests: [insert competing interests here]*	The authors have declared that no competing interests exist.

* typeset	
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Ethics Statement Enter an ethics statement for this submission. This statement is required if the study involved: • Human participants • Human specimens or tissue • Vertebrate animals or cephalopods • Vertebrate embryos or tissues • Field research Write "N/A" if the submission does not require an ethics statement. General guidance is provided below. Consult the submission guidelines for detailed instructions. **Make sure that all information entered here is included in the Methods section of the manuscript.**	N/A

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Data Availability Before publication, Authors are required to make fully available and without restriction all data underlying their findings. Sharing the data underlying your findings increases the visibility, credibility, and impact of your work, while supporting transparency and future scientific discovery. Please see our PLOS Data	No data was generated by this study. The following existing data sources were used: MNIST Database of Handwritten Digits from the UCI Machine Learning Repository, available via https://doi.org/10.24432/C53K8Q and http://yann.lecun.com/exdb/mnist/ Breast Cancer Semantic Segmentation Dataset (BCSS) from the PathologyDataScience GitHub repository, available via https://doi.org/10.1093/bioinformatics/btz083 and https://github.com/PathologyDataScience/BCSS CheXpert Chest X-ray Dataset from the Stanford Center for Artificial Intelligence in

Policy page for detailed information on this policy. A **Data Availability Statement**, detailing where the data can be accessed, is required at first submission. Insert your Data Availability Statement in the box below. Please see the data reporting section of our submission guidelines for instructions on what you need to include in your Data Availability Statement. Examples of acceptable Data Availability Statements are included here for reference: 1. The minimal data set [and accompanying code, where generated] is available at [repository name] via [DOI hyperlink] 2. No data was generated by this study. The following existing data sources were used: [dataset name] from [dataset location] available via [web address or DOI hyperlink]. 3. For Study Protocols: No datasets were generated or analysed during the current study. All relevant data from this study will be made available upon study completion. *PLOS allows rare exemptions to address legal and ethical concerns. If you have legal or ethical restrictions, please use the box below to detail these in full sentences for the Journal team to consider.* This statement will be typeset if the manuscript is accepted for publication.	Medicine and Imaging (AIMI), available via https://aimi.stanford.edu/datasets/chexpert-chest-x-rays PatchCamelyon (PCam) Benchmark Dataset from Zenodo, available via https://doi.org/10.5281/zenodo.2546921 TissueNet Cell Segmentation Dataset from the DeepCell datasets repository, available via https://doi.org/10.1038/s41587-021-01094-0 and https://datasets.deepcell.org/

Managing Editor
Plos One

12-Jan-2026

Dear Editor:

We are submitting our manuscript entitled “Toward reliable false discovery rate control in classification problems under distribution shift” for consideration for publication in *Plos One* as ***original research article***.

We note that we are affiliated to a Russian University, our institution is not under sanctions and we are not involved in any research related to the Russian military, etc. We will pay APC from a European Union (Hungarian) bank card.

We present our study on controlling false discovery rates (FDR) under data distribution shift in classification problems in the biomedical domain. In this study, we draw attention to a very important but often neglected problem; namely, the performance of deep-learning based classifiers is assumed to be the same in application. However, it often can be much worse than thought due to data distribution shift (a.k.a. covariate shift), data label distribution shift, batch effects caused by some confounding factors, and overfitting. This performance drop of a machine learning system often remains undetected, resulting in risk for people in health care and clinical applications.

In this manuscript, we present a simple, fast and robust method to adjust the test prediction scores so that the latent test null distribution will be more similar to the train null distribution. This will result in more accurate FDR control.

Our method operates in the 1-dimensional prediction score space. Therefore, it is not hindered by high-dimensional space, data sparsity (i.e. curse-of-dimensionality), and correlated features. In addition, our method does not rely on data clustering or graphical causal models.

We tested our method with four public benchmark datasets commonly used in biomedical image analysis: PatchCamelyon (PCam), Chest x-ray (CheXpert), TissueNet, and Breast Cancer Semantic Segmentation (BCSS). With our method, we obtained much more accurate FDR control under data distribution shifts in the test data compared to a recent algorithm, called Learn-then-test, developed by Michael Jordan’s lab from Stanford University.

A declaration of any potential competing interests:

None.

Confirmation that all authors have approved the manuscript for submission:

This manuscript has been approved by both authors.

Confirmation that the content of the manuscript has not been published:

This manuscript has never been submitted to any other journals. We have not interacted with PLOS regarding this manuscript before. The manuscript will be properly formatted for the journal if it is needed for the reviewing stage or once it is accepted. We did not use AI chat bots in manuscript preparation or in this study in general.

Suggested Academic Editors:

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4. Xiaohui Zhang, USA, orcid.org/0000-0002-7990-1594
5. Anybody with a background in machine learning classification, error control, image processing would suffice.

Suggested reviewers:

1. Nikita Moshkov, PhD, Institute of Biochemistry, Biological Research Center. Szeged, Hungary, Deep Learning specialist in biological image analysis, cell segmentation, etc. nikita.moshkov@brc.hu
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Opposed reviewers:

None.

We hope that this manuscript provides a useful and interesting study for the readers of Plos One.

Best regards,

Attila Kertesz-Farkas,
Professor
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Toward reliable false discovery rate control in classification problems under distribution shift

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Abstract

Background: Domain shifts or batch effects can significantly degrade the performance of machine learning-based classifiers in software production without any warning. This can particularly be problematic in medical applications which involve human lives. The shift changes the shape of the prediction score distribution of the test samples compared to that of the training samples; hence, decision boundary re-calibration would be required to conform to the expected performance standards.

Method and Findings: We present a simple, yet robust heuristic method to calculate valid p-values (uniformly distributed) for raw prediction scores of test samples under data shift in general classification problems. Our method is based on an adjustment of the test prediction scores so that the test null distribution (approximated distribution of the prediction scores of negative test samples) aligns to the training null distribution (actual distribution of prediction scores of the negative training samples). We termed it the Test Null Adjustment (TNA) method. Then, empirical p-values of the adjusted test prediction scores can be calculated with respect to the training null distribution to be used in any statistical applications which require test samples accompanied by valid p-values. We demonstrate the usability of our TNA method in FDR control with Benjamini-Hochberg (BH) protocol.

Conclusion: The main advantage of the TNA method is that its application does not require any domain specific knowledge. It works in binary and multi-label classification scenarios, it is fully data-driven, and it operates in the 1-dimensional space of the prediction scores ensuring stable numerical calculations. We have performed four experimental tests on the biomedical image analysis domain to demonstrate that the TNA method is able to mitigate the bias in FDR induced by data shift in real-life applications. TNA is available at `https://github.com/Aborevsky01/TNA`

Introduction

Modern AI applications not only need to increase their discriminative power to reduce the overall number of incorrect predictions, but also need to simultaneously minimize the quantified risks of false positive (FP) and negative (FN) error by an optimal decision threshold calibration while also being able to handle (or detect) data distribution shift [1, 2]. False positive (type I) and false negative (type II) errors can have significantly different costs or impacts on people and society [3]. For instance, a false positive HIV, COVID-19 or cancer diagnosis can cause emotional trauma and the person may undergo invasive treatments with undesirable complications and side effects [4–7] until a more elaborate clinical test establishes the correct diagnosis.

However, a failure to diagnose an infectious disease may lead to a false sense of safety [8], and may put not only their own but also other people's lives in danger [9]. An undetected cancer may either delay or hinder treatment [10], and misdiagnosed human papillomavirus (HPV) may progress to cancer precursor lesions [11, 12].

False discovery rate (FDR) control procedures are plausible techniques to keep the FP rate among the total positive classifications at a certain, user-defined level [13], when the risk of FP prediction is significantly higher than the cost of FN predictions. FDR control became popular with the emergence of high-throughput and shotgun experiments which measured large numbers of distinct variables simultaneously per sample in a cost efficient way. For instance, automatic patch clamp systems employ a deep-learning-based image processing module to automatically detect target cells (e.g. nerve cells in living brain). Then it calibrates the movement of a micropipette to approach the cell for future experiments [14]. Cell morphological profiling in high-throughput imaging assays for studying compound libraries and human diseases [15] relies on automatic cell segmentation and detection often in 3D images [16]. The FDR control here could curb the error in selecting wrong cells for subsequent experiments and feature quantification. Microarrays measure the expression levels of thousands of genes simultaneously and the FDR control allows the selection of promising genes for followup studies [17–19]. Tandem mass spectrometry can be used to quantify and identify thousands of proteins in a complex biological/chemical sample and FDR control makes it possible to choose the most trustworthy protein identifications [20].

The False Selection Rate (FSR) control procedure can restrain the FP and FN errors simultaneously [21]. This approach determines two thresholds: one to make definitive positive predictions, and another one to make definitive negative predictions. Any samples not passing either threshold remain in *indecision*. Therefore, the misclassification error (accuracy) among definitive predictions can be bounded, while a subset of samples without predictions are returned for manual investigation by a human expert or for further data collection [22]. Typically, the FDR and FSR controls, and many other statistical methods heavily rely on valid, i.e. uniformly distributed p-values, while typical deep learning models produce raw discriminative scores or logits for test samples.

In this paper, we are interested in providing valid p-values for a set of discriminative scores $T_s = \{t_1, t_2, \ldots, t_m\}$, where each score t_i is produced by a machine learning classifier for a test instance t_i without knowing its true label. The scores t_is are from a mixture of unknown null and non-null hypotheses; i.e. t_i is generated under one of the two possible distributions: (1) null distribution: $T_i \sim T_0$, and (2) non-null distribution: $T_i \sim T_1$; in a mixture model: $T_i \sim T = \pi_0 T_0 + (1 - \pi_0)T_1$, where π_0 is the prior probability of that a score comes from the null distribution. The null distribution is associated with the negative class, the non-null distribution is associated with the positive class. The null and the non-null distributions as well as the assignments are unknown due to the lack of information on the true class labels. Implicitly or explicitly, but knowing or estimating the null distribution is essential to obtain accurate, valid p-values with respect to it. In certain practical applications it is possible to label some of the data as negative with high confidence. For instance, decoy peptides [20] in tandem mass spectrometry and internal negative control proteins [23] in proteomic profiling serve this purpose. However, this is not the case with general machine learning applications, and we propose to approximate the test null distribution T_0 with the empirical training null $X_0 = \{x_1, x_2, \ldots, x_n\}$, where all x_is prediction scores belong to the negative training class; that is, $T_0 \approx X_0$.

Therefore, an *empirical p-value* (EPV) of a given discriminative score t could be defined as the fraction of the scores of the negative training data that are greater than or equal to t, formally $p_t = (1 + \sum_{x_i \in X_0} \mathbb{1}\{x_i \geq t\})/(n + 1)$ [eq.]. This empirical

p-value is also referred to as conformal p-value [24] or as rank among negative controls (RANC) [23]. The EPVs of the negative test samples are super uniform under the null if the data distribution of the test and training samples are identically and independently distributed. This property is called *exchangeability*. In this case, the Benjamini-Hochberg (BH) procedure controls the FDR at $\pi_0\alpha$ level, where α is a user-defined confidence level. The EPVs have been successfully used in recent methods for outlier detection [24, 25] and in classification [22, 26] under the assumption of exchangeability.

Unfortunately, the validity of p-values and EPVs can be compromised by various problems including data distribution shift (a.k.a. covariate shift), data label distribution shift, batch effects [27] caused by some confounding factors, and overfitting. As a consequence multiple testing procedures that rely on valid p-values may fail to control the relevant statistical errors. In natural sciences, this shift can be attributed including, but not limited, to alterations in laboratory or environmental conditions, change in the reagents or atmospheric ozone levels, as well as to personnel preferences and to variation of technical sources [28]. Various high-throughput instruments produced by different manufacturers may also produce slightly differently distributed data about the same experiments. Quantitative morphological features of cells from multi-channel fluorescence microscopy images may depend on lamp intensities, filter patterns, and dyes [29]. Tandem mass spectrometry data are also affected by differences in experimental protocols and data acquisition conditions, reagent batches, or changes in instrumentation [30, 31]. These impacts may lead to incorrect conclusions about the experiments and to improper clinical drug therapies, etc. Similarly in economics and industry, machine learning-based predictions may also become inaccurate after economic shocks, technological advancement, geopolitical turbulence, and pandemic [32]. For instance, energy consumption prediction needs to be recalibrated after energy price change in order to maintain carbon emission reasonably low [33]; credit risk assessment needs to be adapted to monetary policy tightening, increased government expenditure, and income distribution shift caused by inflation [34–36].

Statistically speaking, the data shift violates the exchangeability assumption resulting in different test and training null hypotheses ($T_0 \not\approx X_0$) that further results in invalid p-values and conservatively or liberally (anti-conservatively) biased FDR control. Recent methods, which deal with data shift, aim to (1) detect data shift [37], (2) identify features in tabular data, which cause data shift [38], (3) detect and correct label shift [39], (4) provide explainability about data shift [40, 41], and (5) aim to adapt to data shifts [42–44], respectively. Most of these methods operate in the feature space and rely on data clustering or graphical causal models. They are also challenged by high-dimensional space, data sparsity (i.e. curse-of-dimensionality) [45], and correlated features.

In this paper, we propose a method to calculate EPVs for raw test scores under data or class distribution shift in general classification problems. These EPVs can be used in any statistical applications which rely on valid p-values. We demonstrate the usability of our method in FDR control with BH procedure. Our method, first, approximates the unknown test null distribution T_0 relying on the known training null and non-null distributions. Then, our method adjusts the test prediction scores so that the adjusted test null distribution $\hat{T_0}$ aligns well to the training null distribution; that is, $\hat{T_0} \approx X_0$. Hence, we termed our method Test Null Adjustment (TNA). Fig 1 illustrates this idea. The empirical p-values of the adjusted test scores can be calculated with respect to the training null distribution X_0 with eq [1]. The test null adjusted empirical p-values (TNA EPVs) can then be used by the user in any statistical methods that require valid p-values, including π_0 estimation, or FDR control with any procedure. Valid p-values should be uniformly distributed over the range of $[0, 1]$. We propose to visually monitor

A B

Fig 1. Sketch of TNA. (A) The plot illustrates the distribution shift of the discriminative scores of the test (blue) instances from the training (red) instances. The shaded areas illustrate the observed training and unobserved test null distributions, respectively. The black arrow indicates that the test scores should be shifted to leftward. The TNA method aims to approximate the unknown test null distribution (blue shaded area) based on the known true training null distribution (red shaded area). (B) TNA aims to perform a test score adjustment so that the test null aligns to training null distributions better.

the uniformity of the p-values with Q-Q plots, in which the p-values are plotted against their ordered normalized positions. Any deviation from the diagonal line of the Q-Q plot would indicate a data shift.

We emphasize that the multiplicity addressed throughout this paper is across the m test instances $t_1, \ldots, t_m$, each associated with its own null hypothesis ("instance t_i belongs to the negative class"); it is not a feature-selection problem, in which the multiplicity would instead be across a set of candidate features or variables. Each score t_i is a single, already-computed scalar value produced by a machine learning classifier for one test instance, not a property of an input feature, and the classifier's input features play no further role once this score is obtained.

The TNA method operates solely at the space of prediction score, which is typically one dimensional for each class. Therefore, TNA is not hindered by the curse-of-dimensionality [45]. It can be employed with any machine learning model, given that the classification is based on discriminative scores. This includes deep neural networks, black box models, support vector machines [46], with any kind of data such as tabular, text, audio, image, graph data. The TNA method is data driven as it does not rely on any class of analytical distributions. The only assumption we have for the TNA method is that the score distributions "somewhat resemble" the ones in Fig 2A or in Fig 10 that are typical.

Materials and methods

Preliminaries

Let us suppose that we are given domain D such as images, text, speech, graphs, time series, tabular data, or the mix of these; furthermore, let $X \subseteq D$ and $T \subseteq D$ denote the training and the test data samples, respectively. Let $Y_t \in \{0, 1\}$ denote the true class label of sample $t \in D$. Let $f : D \to R$ be an appropriately trained binary discriminator (classifier), such as a deep neural network, to predict positive instances, where R denotes the real numbers. Without loss of generality, we assume that higher score $f(t)$ indicates a stronger membership to the positive class. The empirical training null distribution is constructed as $X_0 = \{f(t) : Y_t = 0,\ t \in X\}$ from the scores of negative instances, the empirical training target distribution is constructed as $X_1 = \{f(t) : Y_t = 1,\ t \in X\}$ from the scores of positive instances, and the training score set is $X_s = \{f(t) : t \in X\}$. The true test null T_0, and true test target T_1 are unobserved, but the observed test score set is $T_s = \{f(t) : t \in T\}$. We say that we trust (or accept) a sample t predicted as positive if and only if $f(t) > s_t$ for a given decision threshold s_t and we make indecision for samples if $f(t) < s_t$. The false discovery proportion (FDP) in a set of samples $K = \{t \in D\}$ at a score threshold s_t is calculated as

$$FDP(s_t, K) = \frac{1 + |\{t \in K : Y_t = 0,\ f(t) \geq s_t\}|}{|\{t \in K : f(t) \geq s_t\}|},$$

where |.| denotes set cardinality. We reserve FDR for the statistical expectation of FDP. FDR control methods aim to calibrate the threshold s_t so that the error rate among trusted predictions is at most level of α.

We emphasize that the multiplicity being controlled throughout this paper is across test *instances*, not across candidate features. TNA calibrates a single, already-computed scalar discriminative score $f(t)$ per test instance $t \in T$; it does not operate on, select among, or make any claim about the input features consumed by the classifier f. The BH procedure is applied to the m per-instance p-values obtained from the score set $T_s = \{f(t) : t \in T\}$, where the null hypothesis associated with each instance t is "t belongs to the negative class"; this is the classical multiple-testing setting in which the enumeration is over test instances rather than over candidate features. In the multi-label BCSS experiment, this scope extends naturally to (pixel, class) pairs (Results, BCSS subsection), but the unit of multiplicity remains the instance (pixel) classification, not the feature.

The empirical p-values (EPVs) of the test samples are calculated using eq. with respect to X_0. The p-value calculation can be time consuming for large X_0 sets, in this case, a binned, empirical cumulative distribution function of X_0 could be used to calculate the p-values accordingly in an accelerated way. Then, the FDR of a set T can be controlled using the BH protocol and the EPVs at a predetermined α level. For the sake of simplicity, we replace the α with $\alpha/\hat{\pi_0}$ inside BH algorithm, where $\hat{\pi_0}$ is an estimation of π_0 introduced by Storey [47] in the following way:
$\hat{\pi_0} = (1 + \sum_{i=1}^{m} \mathbb{1}\{p_i \geq \lambda_S\})/(m(1 - \lambda_S))$ for $\lambda_S \in (0, 1)$. The q-value of a test sample $t \in T$ is defined as $q_t = \min_{z \leq f(t)} FDP(z, T)$; that is the minimum α level when the sample t would become a trusted positive prediction. The smaller the q-value of a test sample t is, the more trusted positive prediction t is. Note that, the q-value of a sample depends not only on its prediction score, but also on the other samples in the set.

Learn-then-test (LTT)

We acquired the LTT code from its authors' GitHub repository. We specified the following hyperparameters as following: 100 uniformly distributed queried levels of FDR control are iteratively being processed, starting from the 1e-3, with the most appropriate threshold λ_L being selected among the 300 potential threshold values for each iteration. The remainder of the methodology is entirely dependent on the published pipeline, including the comprehensive calculation of the p-values for each threshold during each alpha level test, with a further Bonferroni procedure employed to select the most appropriate border. The final value of the test false positive rate is equal to the number of samples with higher scores than the selected λ_L.

Example 1

We used the well-known MNIST dataset for demonstration [48] (see Table 1). It contains 60K training and 10K test images of 28x28 grey-scaled pixels in 10 classes. There is no data or label shift between the training and test data. We trained a convolutional neural network to classify the digit '1' as target (positive) against all the other digits (negative) (binary classification for the sake of simplicity). Our ConvNet model contained two convolutional layers consisting of 8 kernels with size of 3×3 each, and it was trained with an Adam optimizer with a learning rate of 0.01 for three epochs. Our trained model achieved a 99% accuracy on the test set.

The relevant results are presented in Fig 2. Fig 2A shows the training and test null and target distributions. Fig 2B shows a Q-Q plot of the test EPVs against the theoretical uniform distribution; that is the scatter plot of the EPV against its normalized rank in our case. The dots line up along the diagonal line which shows that

Fig 2. FDR control in MNIST with EPVs. (A) The discriminative score distributions. (B) The Q-Q plot of the EPVs of the test samples against the theoretical uniform distribution. (C) The number of the accepted (trusted) positive predictions at various q-values calculated with true labels (black solid line), BH with EPVs (red dashed line), and LTT (blue dotted line). (D) Deviation between the true and the estimated FDR produced by BH with EPV (red line) and LTT (blue dots), respectively.

the EPVs calculated are uniformly distributed and, consequently, valid. This also indicates that there is no shift between the training and test distributions. Fig 2C shows the number of accepted (trusted) predictions as positives at various q-values (α level of FDR). The ground truth (black solid line) was calculated using the true test labels. Any FDR controlling method (a) yielding more trusted predictions than the ground truth is implied to be liberally biased, and (b) yielding less trusted predictions is conservatively biased. The BH procedure with EPVs (red dashed line) yields closely the same number of trusted predictions at various FDR levels of α (calculated without using the true test labels). This repeatedly indicates that the EPVs are uniform and the BH procedure is an unbiased method. The LTT method also accurately controls the FDR at any α levels (blue dotted line). Finally, Fig 2D shows that the deviation of the estimated FDPs from the true FDPs is small.

TNA protocol

When the empirical training null distribution X_0 is different from the unobserved, true test null distribution T_0, the EPVs will not be uniformly distributed and the BH procedure results in liberal or conservative bias in FDR estimation. In the following steps, we describe two versions of TNA methods that both adjusts the prediction scores so that the approximated test null distribution $\hat{T}_0$ aligns to X_0. The first variation, called TNA- (TNA minus), relies on the assumption that the shapes of the training null (X_0) and test null (T_0) distributions are "similar", but there is no assumption on the form of the non-null distributions. The second variation, called TNA+ (TNA plus), relies on that the shapes of the training non-null (X_1) and test non-null (T_1) distributions are "similar", but there is no assumption on the analytical form of the null distributions. By "similar" distributions we mean that they are close in a statistical sense, such as having a small Wasserstein distance of a Jensen-Shannon divergence; intuitively, their histograms can be aligned without major reshaping. TNA's performance is robust to minor distributional shifts in practice, but it degrades gracefully as this distance increases. The proportion of the nulls may freely vary between the training and test data in the case of both TNA methods, there is no assumption on this. The two approaches differ only at the 3rd step of the following algorithm.

1. Get the density histograms of X_0, X_1, and T_s distributions with identical bin borders and denote them as $\overline{X}_0$, $\overline{X}_1$, and $\overline{T}_s$, respectively. The $\overline{X}[i]$ denotes the value of bin i. We used 100 bins in our experiments.
2. Let $\hat{\pi}_{0_T}$ be an estimation of the proportion of the negative test predictions calculated as follows: $\hat{\pi}_{0_T} = N/(N+P)$, where N and P denotes the total number of the negative and positive predictions in the given test data, respectively, as predicted by the classifier f. We note that this $\hat{\pi}_{0_T}$ estimation can be done with raw discriminative scores and it does not require valid p-values.

3a. TNA-: We create a new density histogram $\hat{T}_0$ that will be an approximation of the test null T_0. The value of the histogram bin i is calculated by $\hat{T}_0[i] = \overline{T}_s[i] \cdot c[i]$, where $c[i] = X_0[i]/X_s[i] \cdot \hat{\pi}_{0_T}/\pi_{0_X}$ and π_{0_X} is the actual proportion of the null samples in the training data calculated with using the true class labels.

3b. TNA+: This approach relies on the following general formulation: $A = (\pi_0) \cdot A_0 + (1 - \pi_0) \cdot A_1$; therefore, the null distribution can be expressed as $A_0 = (A - (1 - \pi_0) \cdot A_1)/\pi_0$. We create a new density histogram $\hat{T}_0$ that will be an approximation of the test null T_0. The value of the histogram bin i is calculated by $\hat{T}_0[i] = (\overline{T}_s[i] - (1 - \hat{\pi}_{0_T})\hat{T}_1[i])/\hat{\pi}_{0_T}$. The unknown $\hat{T}_1$ is approximated by T_s scaled with the proportion of the target calculated from the training data; that is:

$$\hat{T}_1[i] = \overline{T_s}[i] \cdot \frac{\overline{X_1}[i]}{\hat{\pi}_{0_T}\overline{X_0}[i] + (1 - \hat{\pi}_{0_T})\overline{X_1}[i]}.$$

Note that the empirical *training* null and target distributions (X_0, X_1) are weighted with *test* class proportion $\hat{\pi}_{0_T}$. The outcome of division by zero is set to zero.

4. Let μ_{X_0}, σ_{X_0}, $\mu_{\hat{T_0}}$, and $\sigma_{\hat{T_0}}$ are the mean and std of the X_0 and the $\hat{T}_0$ distributions, respectively.
5. Adjust all test scores $t_s \in T_s$ as follows:

$$\hat{t_s} = (t_s - \mu_{\hat{T_0}})/\sigma_{\hat{T_0}} \cdot \sigma_{X_0} + \mu_{X_0}.$$

Finally, calculate the TNA EPVs for each adjusted $\hat{t_s}$ test score with respect to the empirical training null (X_0) with using Eq. [1]. The TNA EPVs then can be used in any statistical applications that rely on valid p-values. In our experiments in the following sections, we demonstrate the usability of the TNA EPVs in FDR control with BH procedure. With TNA EPVs, we re-estimate $\hat{\pi_0}$ for the test instances with the Storey method with $\lambda_S = 0.8$, and run BH procedure to control the FDR at a desired level. The TNA code is publicly available at `https://github.com/Aborevsky01/TNA`.

We note that the actual number of bins or the size of bin-width in step 1 may depend on the range and/or the number of the discriminative scores. Essentially, there should be enough data in each bins to ensure accurate estimations. This might require the analyst to manually adjust the bins in practice. As a practical rule of thumb, we suggest starting from a bin width such that each bin contains at least a few dozen training samples, and coarsening the bins if the resulting density histograms appear noisy.

We also note that we tested a one-dimensional optimal transportation (OT) to adjust training null to test null. Unfortunately, this approach is very sensitive to the test null approximation especially around the decision boundary. In our experiment, OT severely overfitted, while TNA remained more robust to poor tail approximation of the null distributions; consequently, we did not pursue OT further as a baseline in this study.

(i) Data distribution shift:
(ii) Class distribution shift:
A B C D

Fig 3. FDR control under data (top row) and label (bottom row) shift. (A) Q-Q plot of the EPVs (red dots), the TNA- EPVs (green dots) and the TNA+ EPVs (purple dots) against the theoretical uniform distribution (normalized rank). (B) The number of accepted classifications as a function of the Q-values over a critical range (0-0.1) obtained with (i) ground truth (black line), (ii) BH with EPV (red), (iii) BH with TNA- EPV (green), (iv) BH with TNA- EPV (purple), and (v) LTT (blue). (C) Same as (B) but over the entire q-value range. (D) Deviation of the estimated FDP from the true FDP obtained with (i) BH with EPV (red line), (ii) BH with TNA- EPV (green line), (iii) BH with TNA+ EPV (purple line), and (iv) LTT (blue line).

Example 2

To illustrate the case of data distribution shift, we reduced the pixel intensity by 10% of the test images in the MNIST dataset (but training images remained the same). The accuracy of the CNN classifier (the same as trained in Example) remained at 99 %. The evaluation plots are shown in Fig 3(i). The Q-Q plot (Fig. 3(i)A) shows that the test EPVs (red) became biased due to data distribution shift; however, both TNA methods can correct the EPVs yielding uniformly distributed p-values (green, purple). The number of accepted predictions ((Fig. 3(i)BC) obtained with BH with EPVs (red) differs from the ground truth substantially (black line), it became liberally biased. The LTT yielded similar amount of accepted predictions (blue) compared to the ground truth in the critical range of q-values ((Fig. 3(i)B); however, LTT also becomes liberally biased at lower ranges of q-values ((Fig. 3(i)C). The number of accepted predictions obtained with BH with both TNA EPVs (green, purple) remained close to the ground truth over both the critical and lower ranges of q-values implying accurate FDR control under data distribution shift. The actual deviation of the estimated FDP from the true FDP is shown in Fig 3(i)D for each method. The TNA+ is slightly less accurate than TNA- in this test scenario.

Example 3

The FDR estimation is sensitive to changes in class proportions. To illustrate the case of class distribution shift, we resampled the test data so that the number of true negative and positive instances became equal. This approach was used in other articles, e.g. [49]. The accuracy of the CNN classifier (the same as trained in Example) remained at 99 %. The evaluation plots are shown in Fig 3(ii). The Q-Q plot (Fig. 3(ii)A) shows that the EPVs (red) remained unbiased and so did TNA EPVs (green, purple). This is in fact expected, because the shift in class distributions should not affect the actual null distributions; Therefore, the results of BH-based methods rely on the $\hat{\pi_0}$ estimation. BH with EPV and TNA EPV remained accurate and produced results close to the ground truth (red, green, and purple lines in Fig 3(ii)BCD) indicating successful $\hat{\pi_0}$ estimation. The BH procedure is robust to changes in class proportions because it can involve the (re-)estimation of the negative data proportion ($\hat{\pi_0}$) that can be done with, e.g. Storey method. The LTT is also sensitive to changes in class proportions because it relies on the class proportions that are calculated with the training (validation) datasets, and it assumes that class proportions remain unchanged in test phase. Here, LTT resulted in conservative FDR control even at the critical range of q-values (blue in Fig. 3(ii)BC). The bias in FDR produced by LTT (blue) is also visible in Fig 3(ii)D.

Results

The datasets analyzed in this study are listed in Table 1.

The implementation code used to produce the results reported in this manuscript is publicly available at `https://github.com/Aborevsky01/TNA`.

We qualified the stability of the methods with bootstrap analysis. For each dataset, we resampled both the training and the test sets 100 times with replacement, and recomputed the standard EPV, the TNA, and the LTT methods on each resampled set. The resulting variation is indicated on the plots as a band width around each curve. Datasets with a large number of samples, such as TissueNet, and BCSS, show a very small, hardly visible bandwidth on the curves below, reflecting the low sampling variability that comes with a large sample size. The PCam and CheXpert datasets have fewer samples, and consequently shows a visibly wider bandwidth resulting from this variation.

Table 1. Data Availability Statement containing dataset names, repositories, and accession details.

Dataset	Full Name	Accession / URL
MNIST	MNIST Database of Handwritten Digits	DOI: 10.24432/C53K8Q; `http://yann.lecun.com/exdb/mnist/`
BCSS	Breast Cancer Semantic Segmentation Dataset	DOI: 10.1093/bioinformatics/btz083; `https://github.com/PathologyDataScience/BCSS`
CheXpert	CheXpert Chest X-ray Dataset	`https://aimi.stanford.edu/datasets/chexpert-chest-x-rays`
PatchCamelyon	PatchCamelyon (PCam) Benchmark Dataset	DOI: 10.5281/zenodo.2546921
TissueNet	TissueNet Cell Segmentation Dataset	DOI: 10.1038/s41587-021-01094-0; `https://datasets.deepcell.org/`

PCam: tumor classification

The PatchCamelyon (PCam) benchmark dataset is a collection of histopathologic scans of lymph node sections [50] derived from the Camelyon 16 challenge [51] (see Table 1). PCam consists of 327,680 colored images of size 96x96 each. An arbitrarily chosen sample data is shown in Fig 4. The challenge is to identify the presence (positive) or absence (negative) of any histopathology in one image. The dataset creators of PCam emphasize that all the positive and negative instances in the training, validation, test sets are equally distributed, resulting in a class balance of 1:1. We used the pre-trained resnet34-pcam model, which is a popular, deep residual network from the TIA (Tissue Image Analysis) toolbox [52] with an overall f1-score of 0.889, as stated by the authors. We evaluated the uniformity of the p-values of the test samples and the performance of the FDR controlling methods. The results are shown in the first row of Fig 5. The Q-Q plots (Fig. 5-(i)-A) of the test EPVs (red dots) indicate that the test p-values are slightly biased, possibly indicating a slight overfitting and/or distribution shift between the training and test datasets. However, the Q-Q plots of the test EPVs adjusted with TNA- (green dots) and TNA+ (purple dots) show a more uniformly distributed p-values. The plots on Fig. 5-(i)-BCD indicate that the LTT and standard EPV methods yields mostly accurate FDR control with this dataset especially at critical α levels (0-0.1). The TNA methods (both TNA- and TNA+) yield p-values closer to uniform (Fig. 5-(i)-A), yet they result in conservative FDR control due to poor $\hat{\pi}_0$ estimation: the true $\pi_0 = 0.5$, while the Storey method estimated $\hat{\pi}_0 = 0.58$ for the standard EPVs, but $\hat{\pi}_0 = 0.65$ for both TNA- and TNA+.

Fig 4. Instances from PCam dataset.

In our opinion, the actual class proportions in real-life applications can be significantly different from 1:1, making the FDR control with methods like LTT inaccurate in practical applications. We manually resampled the test data so that the positive and negative class balance became 31%:69%, following the original class balance of the Camelyon 16 dataset. The evaluation plots are shown in the second row of Fig 5. The Q-Q plot (Fig. 5-(ii)-A) reveals that the EPVs (red dots) become slightly biased; however, our TNA methods manage to reduce this bias from the EPVs (green dots). The LTT method became liberal in FDR control, the BH protocol with standard EPVs resulted in conservative FDR control; whereas, the BH with the TNA EPVs remained accurate, especially at critical range of q-values (0-0.1) (Fig. 5-(ii)-C).

The bootstrap analysis shows moderately little variation for the PCam dataset, reflecting its relatively large sample size.

(i) Standard PCam dataset:
(ii) PCam with class distribution shift:
(iii) Standard CheXpert dataset:
(iv) CheXpert dataset with class distribution shift:
A B C D

Fig 5. Performance evaluation of FDR controlling methods. See the caption of Fig 3 for plot descriptions. The bandwidth around the black ground truth line indicating the variation from bootstrapping is indicated with a gray dotted.

Fig 6. Instances of CheXpert dataset.

CheXpert: chest x-ray dataset

The CheXpert dataset (see Table 1) contains a total of 224,316 radiographs of both frontal and lateral views of the thorax from ∼65K patients in order to facilitate the automated interpretation of chest X-rays for multi-label classification tasks (i.e. multiple labels can be predicted for each instances) with a total of 14 labels [53]. An arbitrarily chosen sample data is shown in Fig 6. We used the deep neural network model, called DenseNet121, from the `Torchxrayvision` package [54, 55], specifically developed for this CheXpert dataset. For the sake of simplicity, we turned the CheXpert dataset into a binary classification task by treating all instances of pleural effusion as positive and all the others as negative. We selected this label because this was the most prevalent disease in the dataset. The proportion of the positive class became 27 % and 23 % in the training and test datasets, respectively. The DenseNet121 model achieved accuracies of 89 % and 84 % on the training and test sets, respectively.

We evaluated the four FDR controlling methods and plotted the results on Fig 5-(iii). Interestingly, the most confident predictions of the DenseNet121 model in the test set are wrong. This can be seen by the flat region of the ground truth evaluation (black line) at low q-values around 0-0.05 (Fig, 5-(iii)-C). Fortunately, all four FDR controlling methods under investigations managed to capture this and remain close to the ground truth (Fig, 5-(iii)-B-D). The standard EPVs (red dots) tend to be off at the critical range (p-value < 0.1), but this does not affect the FDR controlling results in this benchmark, perhaps because of the confident incorrect predictions. Overall, we conclude that all four methods provide accurate FDR control. In our opinion, this indicates the absence of any database distribution shits. These plots also show that our TNA- and TNA+ algorithms do not corrupt the EPVs in the absence of data shift.

The actual frequency of pleural effusion may be different among all chest x-ray radiographs performed. It may vary over locations, age groups, in-patients and out-patients [56, 57]. In the United States, approximately 275 million conventional radiology procedures are performed annually [58], but only around 1.5 million of US citizens develop pleural effusion [57]. Pleural effusion develops in about 20-40% of in-patient with pneumonia [59]. Whatever the real abundance of the pleural effusion is among chest x-rays performed, it can be significantly different from the 27 % of the CheXpert dataset. Taking this into account, we re-sampled the test set in the CheXpert dataset to achieve a lower positive class ratio of 13%. The results of the re-evaluation are shown in Fig 5-(iv). The LTT and the TNA- EPVs are both slightly biased. The BH with the standard EVPs and with TNA+ EPVs achieve the most accurate FDR control over the entire range of α levels (0-1) in general. The LTT method is very

sensitive to class distribution shift demonstrated by this case, and it resulted in a liberally biased FDR control.

The CheXpert dataset has the smallest test set, and accordingly it shows the highest variability, both in the ground truth and in the estimated curves, as indicated by the bootstrap band widths on the figures.

TissueNet: cells segmentation

Fig 7. Instance of TissueNet dataset. Two-channel original images are being processed by the model, in turn producing masks & flows to correctly segment cells

TissueNet dataset contains 1.3 million microscopic images of cells obtained from six various platforms and nine organs including both histologically healthy and diseased tissues [60](see Table 1). Each images has size of 512x512 pixels, was manually segmented at the pixel level and paired whole-cell and nuclear annotations. An arbitrarily chosen sample data is shown in Fig 7. Our binary classification task was to predict whether a pixel is inside a cell (positive) or not (negative). The overall positive-negative ratio was 59%:41% in the training data and in contrast to 66%:34% in the test data. However, the class ratio varies over the images considerably. We used the Cellpose pre-trained deep neural network model [61] and fine-tuned it for five epochs. It achieved accuracy of 95% and 94% on training and test data, respectively. Our evaluation is reported in Fig 9-(i). The plots reveal that the BH protocol with standard EPVs becomes liberally biased (Fig. 9-(i)-B-C). The latter possibly originates from the data distribution shift or from overfitting. Both LTT and the BH with TNA EPV remain accurate.

The situation changes when we evaluate the segmentation problem with a single image. We arbitrarily chose one test image (ID: 007-008-009) and run the Cellpose model to classify each pixel. The positive and negative class balance was: 33%:67%. The Q-Q plot (Fig. 9-(ii)-A) of the test EPVs (red dots) indicates that the scores of negative samples from training and test datasets are different. The TNA EPVs on the Q-Q plot shows that the TNA methods have managed to adjust the p-values of the test samples (green and purple dots) closer to the uniform distribution. The plots on Fig. 9-(ii)-B-D indicate that LTT has turned out to be liberally biased, the BH method with standard EPVs strongly conservative. The BH with TNA- EPVs (green) and with TNA+ EPVs (purple) remain accurate in the critical region of q-values (0-0.4). Thus, the TNA methods have succeeded in correcting the p-values at the critical range at least.

BCSS: Breast cancer semantic segmentation dataset

Fig 8. An instance from BCSS dataset.

The Breast Cancer Semantic Segmentation (BCSS) dataset [62] contains around 20K image segmentation objects annotated through Digital Slide Archive, with a particular focus on tissue regions derived from breast cancer imagery obtained from The Cancer Genome Atlas (TCGA)(see Table 1). The image pixels are labeled by: 'Stroma' (class id: 0), 'Tumor' (class id: 1) and 'Others' (class id: 2). An arbitrarily chosen sample data is shown in Fig 8. The corresponding class ratios in the training and test sets are 56%:38%:6% and 33%:59%:8%, respectively. We used the deep learning model called `'fcn resnet50 unet-bcss'` from the TIA (Tissue Image Analysis) toolbox [52] to classify the pixels to one of the three classes. The model achieved an accuracy of 84 % and 82 % on the training and test datasets, respectively.

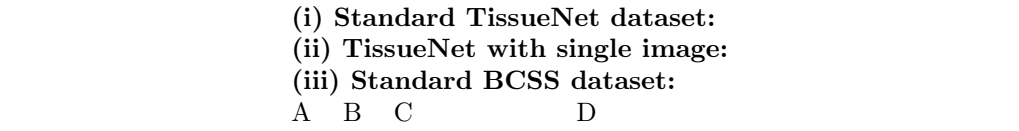


Fig 9. Performance evaluation of FDR controlling methods. See the caption of Fig 3 for plot descriptions. The bootstrap bands are narrow and hardly visible on these curves, due to the large sample size of the TissueNet and BCSS datasets, which results in little sampling variation.

Since this dataset involves a multi-class classification problem in which the FDR control is not straightforward due to the competition nature of the class prediction; thus, we slightly modified the FDR control procedure. We calculated the p-values, the $\hat{\pi_0}$ estimations with Storey method with respect to a single class (deemed as positive); and repeat it for each class via one-vs-rest fashion. Thus, for each pixels, we got three p-values, each corresponding to one of the three classes. Then we concatenated all of these p-values and calculated the q-values. We note the FDR control in this case allows the prediction of a single pixel into multiple classes. This is essentially a multi-label classification scenario.

The evaluation of the FDR controlling methods is shown in Fig 9-(iii). At first glance, the results suggest that LTT and TNA+ manage to control FDR well but this datasets is challenging for BH with standard or TNA- EPVs. The Q-Q plot of the EPVs exhibit deviations from uniformity but TNA EPVs are closer to being uniform (Fig. 9-(iii)-A). Overall, the BH control with standard EPVs is liberal at every alpha level (Fig. 9-(iii)-B); while the other methods are slightly liberal over the critical range of q-values (0-0.1) (Fig. 9-(iii)-C), and they become conservative beyond. BH protocol seems to be more accurate with standard rather than with TNA EPVs. The FDR estimation error with BH is significant (Fig. 9-(iii)-D).

Next, we further analyzed the results by each class separately. We plotted the training and test positive and negative discriminative score distributions class-wise provided by the deep learning model (Fig. 10 first row). The distribution plots show that the test scores are differently distributed from the training scores in all classes. We also plotted the distribution of the adjusted scores of the negative test samples, adjusted by our TNA methods (green and purple lines in Fig. 10) in order to see how well they are aligned to the training null distribution. In the case of the Stroma class, the mode of adjusted test scores (green and purple lines) move a bit closer to the one of the true training null distribution (X0, filled orange bars). However, their tails on the right hand side remain off. In the other two classes, the adjusted test scores align visibly better to the training null distribution (X0). The corresponding FDR curves are shown in the second row of Fig 10. Both the TNA- and the TNA+ methods improves the FDR estimation accuracy with all three classes. This is mainly due to the fact that the Storey method provides better $\hat{\pi_0}$ estimation using the TNA EPVs than using the standard EPVs. The LTT method is overly liberal with the Stroma class, and it is overly conservative with the Tumor class. We suspect that LTT appears to be accurate on the whole dataset simply by chance because the two types of bias cancel each other out in the Stroma and the Tumor classes. These two classes account for the the 94 % of the whole data.

In our opinion, the comparatively weaker performance of the TNA method on the BCSS dataset stems from subpopulation shift (proportions of different subclasses change from training to test sets). The shapes of the modes of the training and test null distributions differ more substantially than in the other benchmark datasets. In such cases, a simple affine adjustment of the test scores cannot adjust the test null mode close to the training null mode. The differing shapes of the modes across the three

classes can be seen in Fig 10. This divergence between the training and test null modes is most pronounced for the Tumor class.

Fig 10. Classwise score distributions and FDR control in the BCSS dataset.

Robustness to binning

Finally, we investigate the robustness of TNA to the hisotgam binning. We run TNA- and TNA+ over various numbers of histogram bins from 5 to 200, and compared the resulting numbers of trusted predictions to the ground truth. Results are shown on Fig 11. The results obtained with 30 bins or more are essentially the same; only the extremely coarse binnings (5–20 bins) deviate noticeably, because the density histograms become too crude to capture the shape of the null. TNA is therefore robust to the binning, and any sufficiently fine binning yields practically the same results.

Fig 11. Robustness to histogram binning. Top row: no shift; bottom row: class distribution shift.

Conclusion

In this study, we examined the effects of data shift on False Discovery Rate (FDR) control. In our opinion, the BH procedure with standard EPVs provides a very accurate FDR control in the absence of any data shift. Unfortunately, data shift is rather common in practice, and we demonstrated that the standard BH and the LTT methods are quite sensitive even to small data or class distribution shift. In this paper, we presented a simple, but robust heuristic method, called Test Null Adjustment (TNA), in order to mitigate the effect of the data shift on the FDR control. Our method operates in the space of prediction scores, thereby it is not affected by the curse-of-dimensionality problem, which often hinders methods that aim to mitigate data shift in the input feature space. The TNA method consists of two main steps. First, it approximates the null distribution among the scores of the test instances via certain statistics from the scores of the training instances. Second, it recalibrates the test scores so that the test and training null distributions would move closer to each other. TNA is fully data-driven, it does not rely on any theoretical assumptions or forms of the data. We have demonstrated in our experimental test with four biomedical datasets that TNA can successfully mitigate the bias caused by data shift; while, leaving the distributions intact in the absence of data shift. We note that, the handling of data shifts in practice is challenging, and unfortunately our method does not provide any theoretical guarantee that the FDR control with TNA EPVs becomes accurate, or conservatively or liberally biased without any additional assumptions; hence TNA is heuristic in its general form.

With this study, we also aim to emphasize and popularize the importance of accurate error control in any predictions in the biomedical domain. We have shown that the deviations in the simple Q-Q plots of the calculated vs. the theoretical p-values can be used to detect data shifts between the training and test data. Moreover, we also hope that the FDR control and plots with q-values will become a standard metric in performance evaluation for prediction systems in the biomedical domain.

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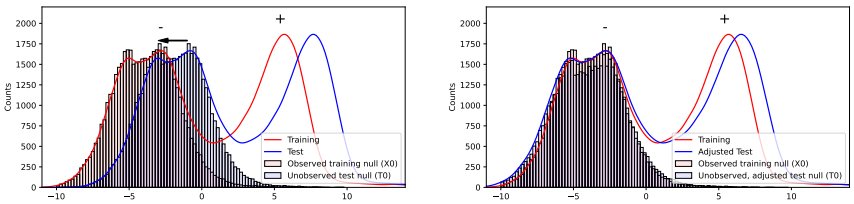


Figure 1: **Sketch of TNA.** (A) The plot illustrates the distribution shift of the discriminative scores of the test (blue) instances from the training (red) instances. The shaded areas illustrate the observed training and unobserved test null distributions, respectively. The black arrow indicates that the test scores should be shifted to leftward. The TNA method aims to approximate the unknown test null distribution (blue shaded area) based on the known true training null distribution (red shaded area). (B) TNA aims to perform a test score adjustment so that the test null aligns to training null distributions better.

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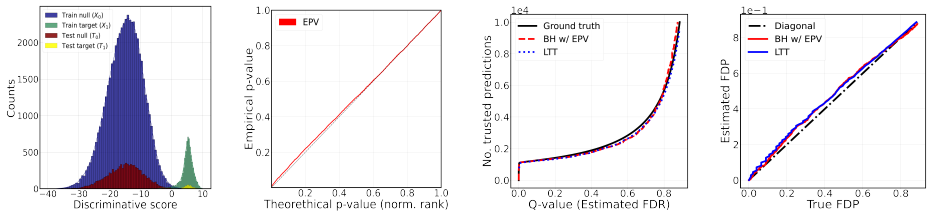


Figure 2: **FDR control in MNIST with EPVs.** (A) The discriminative score distributions. (B) The Q-Q plot of the EPVs of the test samples against the theoretical uniform distribution. (C) The number of the accepted (trusted) positive predictions at various q-values calculated with true labels (black solid line), BH with EPVs (red dashed line), and LTT (blue dotted line). (D) Deviation between the true and the estimated FDR produced by BH with EPV (red line) and LTT (blue dots), respectively.

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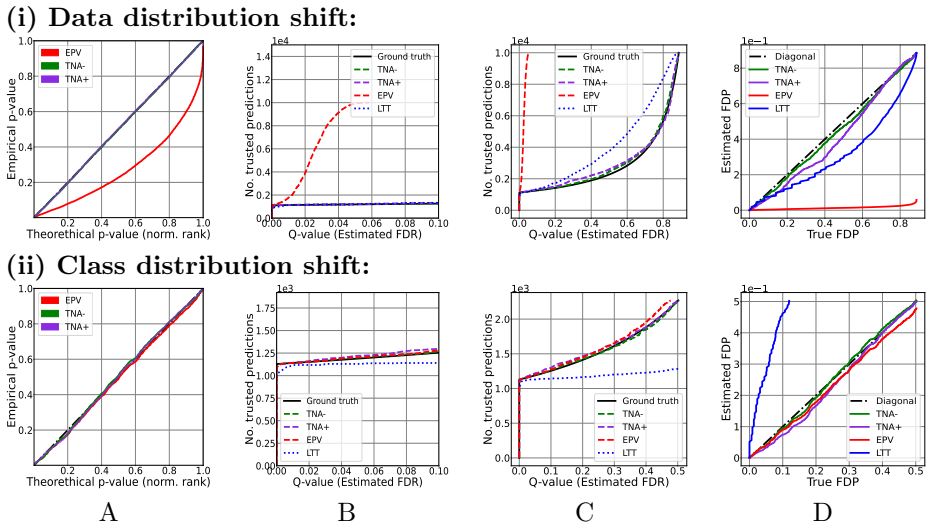


Figure 3: **FDR control under data (top row) and label (bottom row) shift.** (A) Q-Q plot of the EPVs (red dots), the TNA- EPVs (green dots) and the TNA+ EPVs (purple dots) against the theoretical uniform distribution (normalized rank). (B) The number of accepted classifications as a function of the Q-values over a critical range (0-0.1) obtained with (i) ground truth (black line), (ii) BH with EPV (red), (iii) BH with TNA- EPV (green), (iv) BH with TNA- EPV (purple), and (v) LTT (blue). (C) Same as (B) but over the entire q-value range. (D) Deviation of the estimated FDP from the true FDP obtained with (i) BH with EPV (red line), (ii) BH with TNA- EPV (green line), (iii) BH with TNA+ EPV (purple line), and (iv) LTT (blue line).

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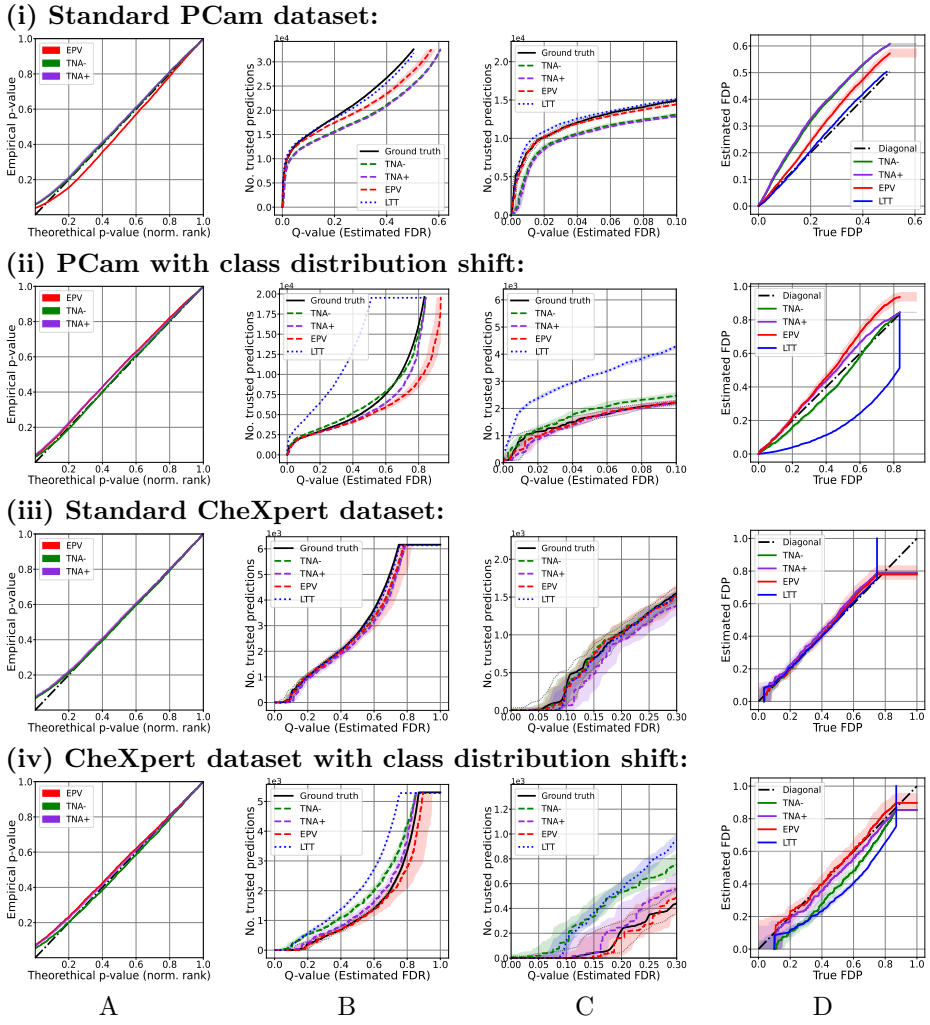


Figure 5: **Performance evaluation of FDR controlling methods.** See the caption of Fig **??** for plot descriptions.

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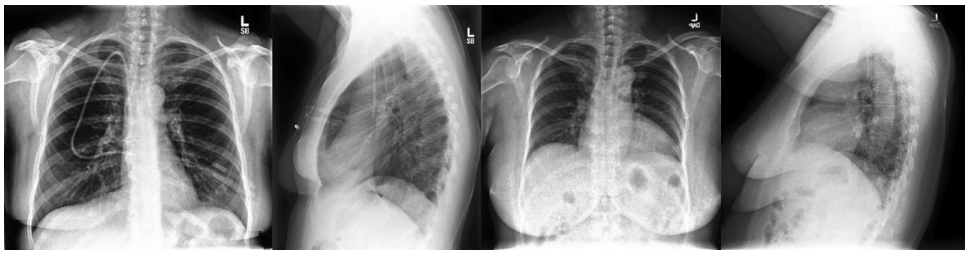


Figure 6: **Instances of CheXpert dataset.**

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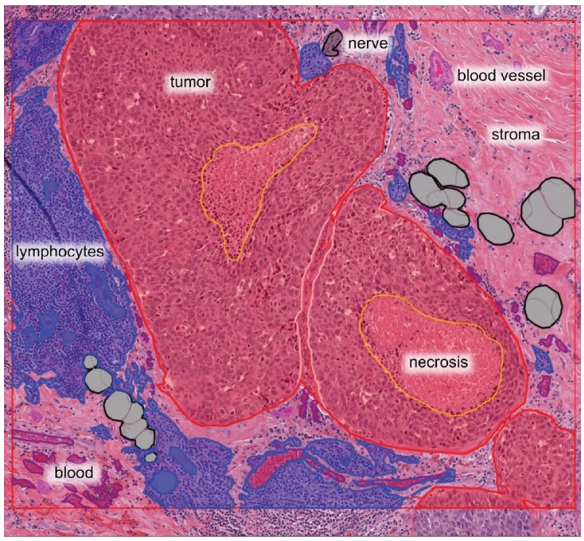


Figure 8: **An instance from BCSS dataset.**

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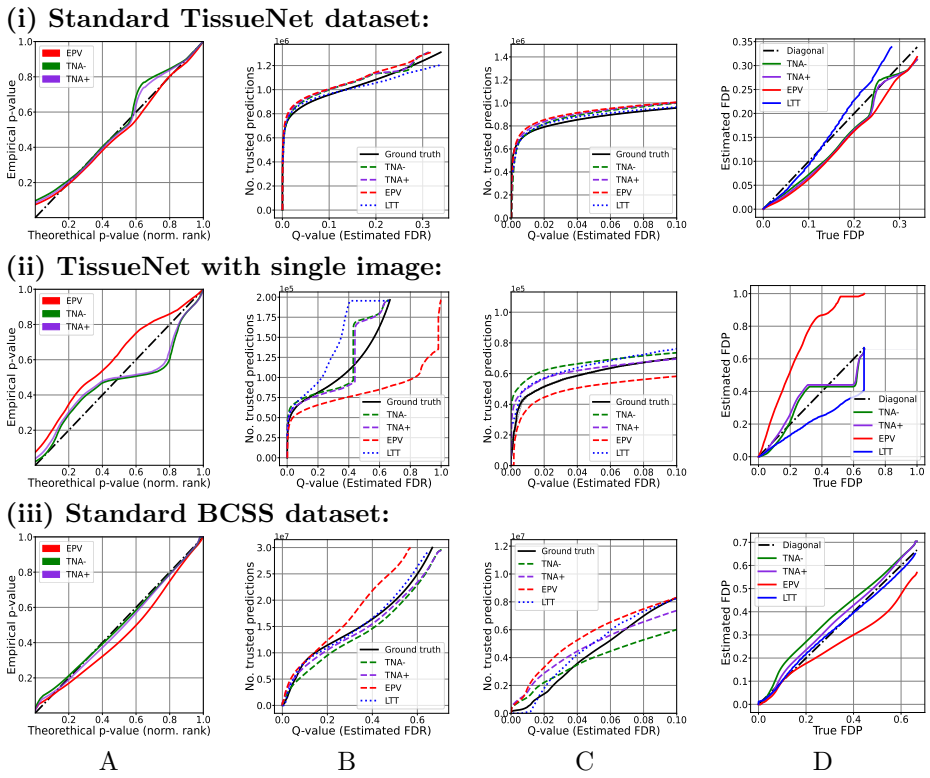


Figure 9: **Performance evaluation of FDR controlling methods.** See the caption of Fig **??** for plot descriptions. The bootstrap bands are narrow and hardly visible on these curves, due to the large sample size of the TissueNet and BCSS datasets, which results in little sampling variation.

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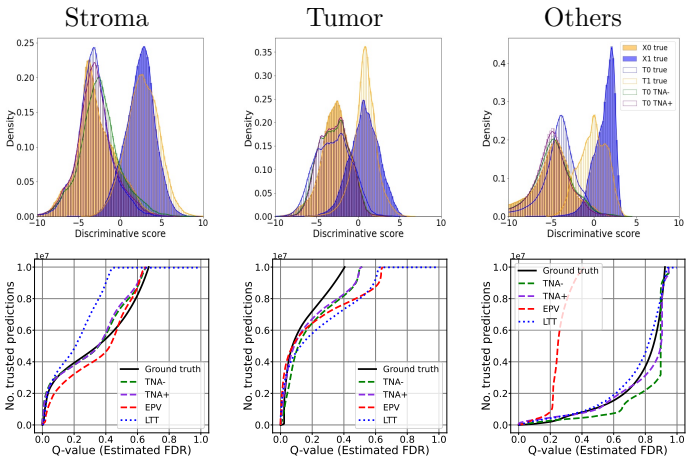


Figure 10: **Classwise score distributions and FDR control in the BCSS dataset.**

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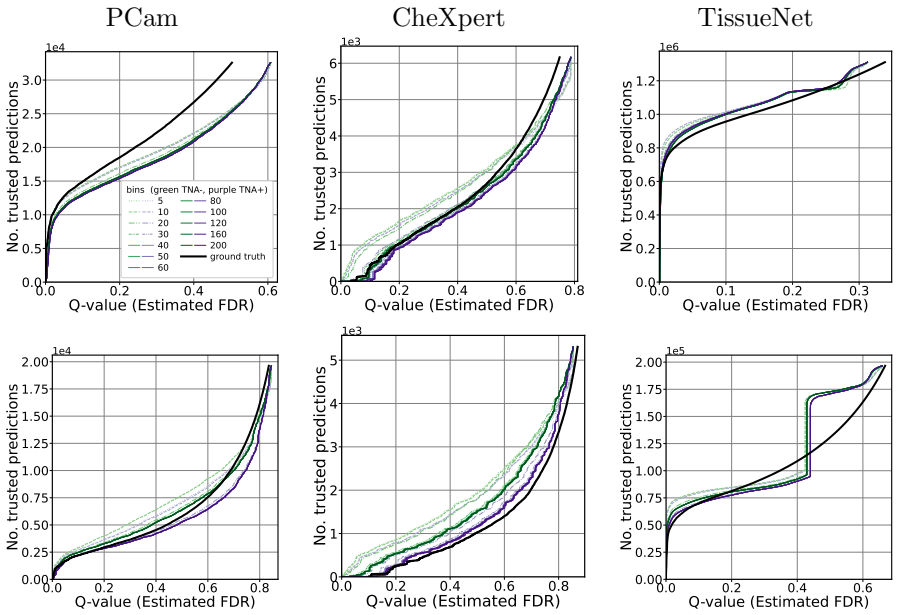


Figure 11: **Robustness to histogram binning.** Top row: no shift; bottom row: class distribution shift.

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Toward reliable false discovery rate control in classification problems under distribution shift

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Abstract

Background: Domain shifts or batch effects can significantly degrade the performance of machine learning-based classifiers in software production without any warning. This can particularly be problematic in medical applications which involve human lives. The shift changes the shape of the prediction score distribution of the test samples compared to that of the training samples; hence, decision boundary re-calibration would be required to conform to the expected performance standards.

Method and Findings: We present a simple, yet robust heuristic method to calculate valid p-values (uniformly distributed) for raw prediction scores of test samples under data shift in general classification problems. Our method is based on an adjustment of the test prediction scores so that the test null distribution (approximated distribution of the prediction scores of negative test samples) aligns to the training null distribution (actual distribution of prediction scores of the negative training samples). We termed it the Test Null Adjustment (TNA) method. Then, empirical p-values of the adjusted test prediction scores can be calculated with respect to the training null distribution to be used in any statistical applications which require test samples accompanied by valid p-values. We demonstrate the usability of our TNA method in FDR control with Benjamini-Hochberg (BH) protocol.

Conclusion: The main advantage of the TNA method is that its application does not require any domain specific knowledge. It works in binary and multi-label classification scenarios, it is fully data-driven, and it operates in the 1-dimensional space of the prediction scores ensuring stable numerical calculations. We have performed four experimental tests on the biomedical image analysis domain to demonstrate that the TNA method is able to mitigate the bias in FDR induced by data shift in real-life applications. TNA is available at `https://github.com/Aborevsky01/TNA`

Introduction

Modern AI applications not only need to increase their discriminative power to reduce the overall number of incorrect predictions, but also need to simultaneously minimize the quantified risks of false positive (FP) and negative (FN) error by an optimal decision threshold calibration while also being able to handle (or detect) data distribution shift [1, 2]. False positive (type I) and false negative (type II) errors can have significantly different costs or impacts on people and society [3]. For instance, a false positive HIV, COVID-19 or cancer diagnosis can cause emotional trauma and the person may undergo invasive treatments with undesirable complications and side effects [4–7] until a more elaborate clinical test establishes the correct diagnosis.

However, a failure to diagnose an infectious disease may lead to a false sense of safety [8], and may put not only their own but also other people's lives in danger [9]. An undetected cancer may either delay or hinder treatment [10], and misdiagnosed human papillomavirus (HPV) may progress to cancer precursor lesions [11, 12].

False discovery rate (FDR) control procedures are plausible techniques to keep the FP rate among the total positive classifications at a certain, user-defined level [13], when the risk of FP prediction is significantly higher than the cost of FN predictions. FDR control became popular with the emergence of high-throughput and shotgun experiments which measured large numbers of distinct variables simultaneously per sample in a cost efficient way. For instance, automatic patch clamp systems employ a deep-learning-based image processing module to automatically detect target cells (e.g. nerve cells in living brain). Then it calibrates the movement of a micropipette to approach the cell for future experiments [14]. Cell morphological profiling in high-throughput imaging assays for studying compound libraries and human diseases [15] relies on automatic cell segmentation and detection often in 3D images [16]. The FDR control here could curb the error in selecting wrong cells for subsequent experiments and feature quantification. Microarrays measure the expression levels of thousands of genes simultaneously and the FDR control allows the selection of promising genes for followup studies [17–19]. Tandem mass spectrometry can be used to quantify and identify thousands of proteins in a complex biological/chemical sample and FDR control makes it possible to choose the most trustworthy protein identifications [20].

The False Selection Rate (FSR) control procedure can restrain the FP and FN errors simultaneously [21]. This approach determines two thresholds: one to make definitive positive predictions, and another one to make definitive negative predictions. Any samples not passing either threshold remain in *indecision*. Therefore, the misclassification error (accuracy) among definitive predictions can be bounded, while a subset of samples without predictions are returned for manual investigation by a human expert or for further data collection [22]. Typically, the FDR and FSR controls, and many other statistical methods heavily rely on valid, i.e. uniformly distributed p-values, while typical deep learning models produce raw discriminative scores or logits for test samples.

In this paper, we are interested in providing valid p-values for a set of discriminative scores $T_s = \{t_1, t_2, \ldots, t_m\}$, where each score t_i is produced by a machine learning classifier for a test instance t_i without knowing its true label. The scores t_is are from a mixture of unknown null and non-null hypotheses; i.e. t_i is generated under one of the two possible distributions: (1) null distribution: $T_i \sim T_0$, and (2) non-null distribution: $T_i \sim T_1$; in a mixture model: $T_i \sim T = \pi_0 T_0 + (1 - \pi_0)T_1$, where π_0 is the prior probability of that a score comes from the null distribution. The null distribution is associated with the negative class, the non-null distribution is associated with the positive class. The null and the non-null distributions as well as the assignments are unknown due to the lack of information on the true class labels. Implicitly or explicitly, but knowing or estimating the null distribution is essential to obtain accurate, valid p-values with respect to it. In certain practical applications it is possible to label some of the data as negative with high confidence. For instance, decoy peptides [20] in tandem mass spectrometry and internal negative control proteins [23] in proteomic profiling serve this purpose. However, this is not the case with general machine learning applications, and we propose to approximate the test null distribution T_0 with the empirical training null $X_0 = \{x_1, x_2, \ldots, x_n\}$, where all x_is prediction scores belong to the negative training class; that is, $T_0 \approx X_0$.

Therefore, an *empirical p-value* (EPV) of a given discriminative score t could be defined as the fraction of the scores of the negative training data that are greater than or equal to t, formally $p_t = (1 + \sum_{x_i \in X_0} \mathbb{1}\{x_i \geq t\})/(n + 1)$ [eq.]. This empirical

p-value is also referred to as conformal p-value [24] or as rank among negative controls (RANC) [23]. The EPVs of the negative test samples are super uniform under the null if the data distribution of the test and training samples are identically and independently distributed. This property is called *exchangeability*. In this case, the Benjamini-Hochberg (BH) procedure controls the FDR at $\pi_0\alpha$ level, where α is a user-defined confidence level. The EPVs have been successfully used in recent methods for outlier detection [24, 25] and in classification [22, 26] under the assumption of exchangeability.

Unfortunately, the validity of p-values and EPVs can be compromised by various problems including data distribution shift (a.k.a. covariate shift), data label distribution shift, batch effects [27] caused by some confounding factors, and overfitting. As a consequence multiple testing procedures that rely on valid p-values may fail to control the relevant statistical errors. In natural sciences, this shift can be attributed including, but not limited, to alterations in laboratory or environmental conditions, change in the reagents or atmospheric ozone levels, as well as to personnel preferences and to variation of technical sources [28]. Various high-throughput instruments produced by different manufacturers may also produce slightly differently distributed data about the same experiments. Quantitative morphological features of cells from multi-channel fluorescence microscopy images may depend on lamp intensities, filter patterns, and dyes [29]. Tandem mass spectrometry data are also affected by differences in experimental protocols and data acquisition conditions, reagent batches, or changes in instrumentation [30, 31]. These impacts may lead to incorrect conclusions about the experiments and to improper clinical drug therapies, etc. Similarly in economics and industry, machine learning-based predictions may also become inaccurate after economic shocks, technological advancement, geopolitical turbulence, and pandemic [32]. For instance, energy consumption prediction needs to be recalibrated after energy price change in order to maintain carbon emission reasonably low [33]; credit risk assessment needs to be adapted to monetary policy tightening, increased government expenditure, and income distribution shift caused by inflation [34–36].

Statistically speaking, the data shift violates the exchangeability assumption resulting in different test and training null hypotheses ($T_0 \not\approx X_0$) that further results in invalid p-values and conservatively or liberally (anti-conservatively) biased FDR control. Recent methods, which deal with data shift, aim to (1) detect data shift [37], (2) identify features in tabular data, which cause data shift [38], (3) detect and correct label shift [39], (4) provide explainability about data shift [40, 41], and (5) aim to adapt to data shifts [42–44], respectively. Most of these methods operate in the feature space and rely on data clustering or graphical causal models. They are also challenged by high-dimensional space, data sparsity (i.e. curse-of-dimensionality) [45], and correlated features.

In this paper, we propose a method to calculate EPVs for raw test scores under data or class distribution shift in general classification problems. These EPVs can be used in any statistical applications which rely on valid p-values. We demonstrate the usability of our method in FDR control with BH procedure. Our method, first, approximates the unknown test null distribution T_0 relying on the known training null and non-null distributions. Then, our method adjusts the test prediction scores so that the adjusted test null distribution $\hat{T_0}$ aligns well to the training null distribution; that is, $\hat{T_0} \approx X_0$. Hence, we termed our method Test Null Adjustment (TNA). Fig 1 illustrates this idea. The empirical p-values of the adjusted test scores can be calculated with respect to the training null distribution X_0 with eq [1]. The test null adjusted empirical p-values (TNA EPVs) can then be used by the user in any statistical methods that require valid p-values, including π_0 estimation, or FDR control with any procedure. Valid p-values should be uniformly distributed over the range of $[0, 1]$. We propose to visually monitor

A B

Fig 1. Sketch of TNA. (A) The plot illustrates the distribution shift of the discriminative scores of the test (blue) instances from the training (red) instances. The shaded areas illustrate the observed training and unobserved test null distributions, respectively. The black arrow indicates that the test scores should be shifted to leftward. The TNA method aims to approximate the unknown test null distribution (blue shaded area) based on the known true training null distribution (red shaded area). (B) TNA aims to perform a test score adjustment so that the test null aligns to training null distributions better.

the uniformity of the p-values with Q-Q plots, in which the p-values are plotted against their ordered normalized positions. Any deviation from the diagonal line of the Q-Q plot would indicate a data shift.

We emphasize that the multiplicity addressed throughout this paper is across the m test instances $t_1, \ldots, t_m$, each associated with its own null hypothesis ("instance t_i belongs to the negative class"); it is not a feature-selection problem, in which the multiplicity would instead be across a set of candidate features or variables. Each score t_i is a single, already-computed scalar value produced by a machine learning classifier for one test instance, not a property of an input feature, and the classifier's input features play no further role once this score is obtained.

The TNA method operates solely at the space of prediction score, which is typically one dimensional for each class. Therefore, TNA is not hindered by the curse-of-dimensionality [45]. It can be employed with any machine learning model, given that the classification is based on discriminative scores. This includes deep neural networks, black box models, support vector machines [46], with any kind of data such as tabular, text, audio, image, graph data. The TNA method is data driven as it does not rely on any class of analytical distributions. The only assumption we have for the TNA method is that the score distributions "somewhat resemble" the ones in Fig 2A or in Fig 10 that are typical.

Materials and methods

Preliminaries

Let us suppose that we are given domain D such as images, text, speech, graphs, time series, tabular data, or the mix of these; furthermore, let $X \subseteq D$ and $T \subseteq D$ denote the training and the test data samples, respectively. Let $Y_t \in \{0, 1\}$ denote the true class label of sample $t \in D$. Let $f : D \to R$ be an appropriately trained binary discriminator (classifier), such as a deep neural network, to predict positive instances, where R denotes the real numbers. Without loss of generality, we assume that higher score $f(t)$ indicates a stronger membership to the positive class. The empirical training null distribution is constructed as $X_0 = \{f(t) : Y_t = 0,\ t \in X\}$ from the scores of negative instances, the empirical training target distribution is constructed as $X_1 = \{f(t) : Y_t = 1,\ t \in X\}$ from the scores of positive instances, and the training score set is $X_s = \{f(t) : t \in X\}$. The true test null T_0, and true test target T_1 are unobserved, but the observed test score set is $T_s = \{f(t) : t \in T\}$. We say that we trust (or accept) a sample t predicted as positive if and only if $f(t) > s_t$ for a given decision threshold s_t and we make indecision for samples if $f(t) < s_t$. The false discovery proportion (FDP) in a set of samples $K = \{t \in D\}$ at a score threshold s_t is calculated as

$$FDP(s_t, K) = \frac{1 + |\{t \in K : Y_t = 0,\ f(t) \geq s_t\}|}{|\{t \in K : f(t) \geq s_t\}|},$$

where |.| denotes set cardinality. We reserve FDR for the statistical expectation of FDP. FDR control methods aim to calibrate the threshold s_t so that the error rate among trusted predictions is at most level of α.

We emphasize that the multiplicity being controlled throughout this paper is across test *instances*, not across candidate features. TNA calibrates a single, already-computed scalar discriminative score $f(t)$ per test instance $t \in T$; it does not operate on, select among, or make any claim about the input features consumed by the classifier f. The BH procedure is applied to the m per-instance p-values obtained from the score set $T_s = \{f(t) : t \in T\}$, where the null hypothesis associated with each instance t is "t belongs to the negative class"; this is the classical multiple-testing setting in which the enumeration is over test instances rather than over candidate features. In the multi-label BCSS experiment, this scope extends naturally to (pixel, class) pairs (Results, BCSS subsection), but the unit of multiplicity remains the instance (pixel) classification, not the feature.

The empirical p-values (EPVs) of the test samples are calculated using eq. with respect to X_0. The p-value calculation can be time consuming for large X_0 sets, in this case, a binned, empirical cumulative distribution function of X_0 could be used to calculate the p-values accordingly in an accelerated way. Then, the FDR of a set T can be controlled using the BH protocol and the EPVs at a predetermined α level. For the sake of simplicity, we replace the α with $\alpha/\hat{\pi_0}$ inside BH algorithm, where $\hat{\pi_0}$ is an estimation of π_0 introduced by Storey [47] in the following way:
$\hat{\pi_0} = (1 + \sum_{i=1}^{m} \mathbb{1}\{p_i \geq \lambda_S\})/(m(1 - \lambda_S))$ for $\lambda_S \in (0, 1)$. The q-value of a test sample $t \in T$ is defined as $q_t = \min_{z \leq f(t)} FDP(z, T)$; that is the minimum α level when the sample t would become a trusted positive prediction. The smaller the q-value of a test sample t is, the more trusted positive prediction t is. Note that, the q-value of a sample depends not only on its prediction score, but also on the other samples in the set.

Learn-then-test (LTT)

We acquired the LTT code from its authors' GitHub repository. We specified the following hyperparameters as following: 100 uniformly distributed queried levels of FDR control are iteratively being processed, starting from the 1e-3, with the most appropriate threshold λ_L being selected among the 300 potential threshold values for each iteration. The remainder of the methodology is entirely dependent on the published pipeline, including the comprehensive calculation of the p-values for each threshold during each alpha level test, with a further Bonferroni procedure employed to select the most appropriate border. The final value of the test false positive rate is equal to the number of samples with higher scores than the selected λ_L.

Example 1

We used the well-known MNIST dataset for demonstration [48] (see Table 1). It contains 60K training and 10K test images of 28x28 grey-scaled pixels in 10 classes. There is no data or label shift between the training and test data. We trained a convolutional neural network to classify the digit '1' as target (positive) against all the other digits (negative) (binary classification for the sake of simplicity). Our ConvNet model contained two convolutional layers consisting of 8 kernels with size of 3×3 each, and it was trained with an Adam optimizer with a learning rate of 0.01 for three epochs. Our trained model achieved a 99% accuracy on the test set.

The relevant results are presented in Fig 2. Fig 2A shows the training and test null and target distributions. Fig 2B shows a Q-Q plot of the test EPVs against the theoretical uniform distribution; that is the scatter plot of the EPV against its normalized rank in our case. The dots line up along the diagonal line which shows that

Fig 2. FDR control in MNIST with EPVs. (A) The discriminative score distributions. (B) The Q-Q plot of the EPVs of the test samples against the theoretical uniform distribution. (C) The number of the accepted (trusted) positive predictions at various q-values calculated with true labels (black solid line), BH with EPVs (red dashed line), and LTT (blue dotted line). (D) Deviation between the true and the estimated FDR produced by BH with EPV (red line) and LTT (blue dots), respectively.

the EPVs calculated are uniformly distributed and, consequently, valid. This also indicates that there is no shift between the training and test distributions. Fig 2C shows the number of accepted (trusted) predictions as positives at various q-values (α level of FDR). The ground truth (black solid line) was calculated using the true test labels. Any FDR controlling method (a) yielding more trusted predictions than the ground truth is implied to be liberally biased, and (b) yielding less trusted predictions is conservatively biased. The BH procedure with EPVs (red dashed line) yields closely the same number of trusted predictions at various FDR levels of α (calculated without using the true test labels). This repeatedly indicates that the EPVs are uniform and the BH procedure is an unbiased method. The LTT method also accurately controls the FDR at any α levels (blue dotted line). Finally, Fig 2D shows that the deviation of the estimated FDPs from the true FDPs is small.

TNA protocol

When the empirical training null distribution X_0 is different from the unobserved, true test null distribution T_0, the EPVs will not be uniformly distributed and the BH procedure results in liberal or conservative bias in FDR estimation. In the following steps, we describe two versions of TNA methods that both adjusts the prediction scores so that the approximated test null distribution $\hat{T}_0$ aligns to X_0. The first variation, called TNA- (TNA minus), relies on the assumption that the shapes of the training null (X_0) and test null (T_0) distributions are "similar", but there is no assumption on the form of the non-null distributions. The second variation, called TNA+ (TNA plus), relies on that the shapes of the training non-null (X_1) and test non-null (T_1) distributions are "similar", but there is no assumption on the analytical form of the null distributions. By "similar" distributions we mean that they are close in a statistical sense, such as having a small Wasserstein distance of a Jensen-Shannon divergence; intuitively, their histograms can be aligned without major reshaping. TNA's performance is robust to minor distributional shifts in practice, but it degrades gracefully as this distance increases. The proportion of the nulls may freely vary between the training and test data in the case of both TNA methods, there is no assumption on this. The two approaches differ only at the 3rd step of the following algorithm.

1. Get the density histograms of X_0, X_1, and T_s distributions with identical bin borders and denote them as $\overline{X}_0$, $\overline{X}_1$, and $\overline{T}_s$, respectively. The $\overline{X}[i]$ denotes the value of bin i. We used 100 bins in our experiments.
2. Let $\hat{\pi}_{0_T}$ be an estimation of the proportion of the negative test predictions calculated as follows: $\hat{\pi}_{0_T} = N/(N+P)$, where N and P denotes the total number of the negative and positive predictions in the given test data, respectively, as predicted by the classifier f. We note that this $\hat{\pi}_{0_T}$ estimation can be done with raw discriminative scores and it does not require valid p-values.

3a. TNA-: We create a new density histogram $\hat{T}_0$ that will be an approximation of the test null T_0. The value of the histogram bin i is calculated by $\hat{T}_0[i] = \overline{T}_s[i] \cdot c[i]$, where $c[i] = X_0[i]/X_s[i] \cdot \hat{\pi}_{0_T}/\pi_{0_X}$ and π_{0_X} is the actual proportion of the null samples in the training data calculated with using the true class labels.

3b. TNA+: This approach relies on the following general formulation: $A = (\pi_0) \cdot A_0 + (1 - \pi_0) \cdot A_1$; therefore, the null distribution can be expressed as $A_0 = (A - (1 - \pi_0) \cdot A_1)/\pi_0$. We create a new density histogram $\hat{T}_0$ that will be an approximation of the test null T_0. The value of the histogram bin i is calculated by $\hat{T}_0[i] = (\overline{T}_s[i] - (1 - \hat{\pi}_{0_T})\hat{T}_1[i])/\hat{\pi}_{0_T}$. The unknown $\hat{T}_1$ is approximated by T_s scaled with the proportion of the target calculated from the training data; that is:

$$\hat{T}_1[i] = \overline{T_s}[i] \cdot \frac{\overline{X_1}[i]}{\hat{\pi}_{0_T}\overline{X_0}[i] + (1 - \hat{\pi}_{0_T})\overline{X_1}[i]}.$$

Note that the empirical *training* null and target distributions (X_0, X_1) are weighted with *test* class proportion $\hat{\pi}_{0_T}$. The outcome of division by zero is set to zero.

4. Let μ_{X_0}, σ_{X_0}, $\mu_{\hat{T_0}}$, and $\sigma_{\hat{T_0}}$ are the mean and std of the X_0 and the $\hat{T}_0$ distributions, respectively.
5. Adjust all test scores $t_s \in T_s$ as follows:

$$\hat{t_s} = (t_s - \mu_{\hat{T_0}})/\sigma_{\hat{T_0}} \cdot \sigma_{X_0} + \mu_{X_0}.$$

Finally, calculate the TNA EPVs for each adjusted $\hat{t_s}$ test score with respect to the empirical training null (X_0) with using Eq. [1]. The TNA EPVs then can be used in any statistical applications that rely on valid p-values. In our experiments in the following sections, we demonstrate the usability of the TNA EPVs in FDR control with BH procedure. With TNA EPVs, we re-estimate $\hat{\pi_0}$ for the test instances with the Storey method with $\lambda_S = 0.8$, and run BH procedure to control the FDR at a desired level. The TNA code is publicly available at `https://github.com/Aborevsky01/TNA`.

We note that the actual number of bins or the size of bin-width in step 1 may depend on the range and/or the number of the discriminative scores. Essentially, there should be enough data in each bins to ensure accurate estimations. This might require the analyst to manually adjust the bins in practice. As a practical rule of thumb, we suggest starting from a bin width such that each bin contains at least a few dozen training samples, and coarsening the bins if the resulting density histograms appear noisy.

We also note that we tested a one-dimensional optimal transportation (OT) to adjust training null to test null. Unfortunately, this approach is very sensitive to the test null approximation especially around the decision boundary. In our experiment, OT severely overfitted, while TNA remained more robust to poor tail approximation of the null distributions; consequently, we did not pursue OT further as a baseline in this study.

(i) Data distribution shift:
(ii) Class distribution shift:
A B C D

Fig 3. FDR control under data (top row) and label (bottom row) shift. (A) Q-Q plot of the EPVs (red dots), the TNA- EPVs (green dots) and the TNA+ EPVs (purple dots) against the theoretical uniform distribution (normalized rank). (B) The number of accepted classifications as a function of the Q-values over a critical range (0-0.1) obtained with (i) ground truth (black line), (ii) BH with EPV (red), (iii) BH with TNA- EPV (green), (iv) BH with TNA- EPV (purple), and (v) LTT (blue). (C) Same as (B) but over the entire q-value range. (D) Deviation of the estimated FDP from the true FDP obtained with (i) BH with EPV (red line), (ii) BH with TNA- EPV (green line), (iii) BH with TNA+ EPV (purple line), and (iv) LTT (blue line).

Example 2

To illustrate the case of data distribution shift, we reduced the pixel intensity by 10% of the test images in the MNIST dataset (but training images remained the same). The accuracy of the CNN classifier (the same as trained in Example) remained at 99 %. The evaluation plots are shown in Fig 3(i). The Q-Q plot (Fig. 3(i)A) shows that the test EPVs (red) became biased due to data distribution shift; however, both TNA methods can correct the EPVs yielding uniformly distributed p-values (green, purple). The number of accepted predictions ((Fig. 3(i)BC) obtained with BH with EPVs (red) differs from the ground truth substantially (black line), it became liberally biased. The LTT yielded similar amount of accepted predictions (blue) compared to the ground truth in the critical range of q-values ((Fig. 3(i)B); however, LTT also becomes liberally biased at lower ranges of q-values ((Fig. 3(i)C). The number of accepted predictions obtained with BH with both TNA EPVs (green, purple) remained close to the ground truth over both the critical and lower ranges of q-values implying accurate FDR control under data distribution shift. The actual deviation of the estimated FDP from the true FDP is shown in Fig 3(i)D for each method. The TNA+ is slightly less accurate than TNA- in this test scenario.

Example 3

The FDR estimation is sensitive to changes in class proportions. To illustrate the case of class distribution shift, we resampled the test data so that the number of true negative and positive instances became equal. This approach was used in other articles, e.g. [49]. The accuracy of the CNN classifier (the same as trained in Example) remained at 99 %. The evaluation plots are shown in Fig 3(ii). The Q-Q plot (Fig. 3(ii)A) shows that the EPVs (red) remained unbiased and so did TNA EPVs (green, purple). This is in fact expected, because the shift in class distributions should not affect the actual null distributions; Therefore, the results of BH-based methods rely on the $\hat{\pi_0}$ estimation. BH with EPV and TNA EPV remained accurate and produced results close to the ground truth (red, green, and purple lines in Fig 3(ii)BCD) indicating successful $\hat{\pi_0}$ estimation. The BH procedure is robust to changes in class proportions because it can involve the (re-)estimation of the negative data proportion ($\hat{\pi_0}$) that can be done with, e.g. Storey method. The LTT is also sensitive to changes in class proportions because it relies on the class proportions that are calculated with the training (validation) datasets, and it assumes that class proportions remain unchanged in test phase. Here, LTT resulted in conservative FDR control even at the critical range of q-values (blue in Fig. 3(ii)BC). The bias in FDR produced by LTT (blue) is also visible in Fig 3(ii)D.

Results

The datasets analyzed in this study are listed in Table 1.

The implementation code used to produce the results reported in this manuscript is publicly available at `https://github.com/Aborevsky01/TNA`.

We qualified the stability of the methods with bootstrap analysis. For each dataset, we resampled both the training and the test sets 100 times with replacement, and recomputed the standard EPV, the TNA, and the LTT methods on each resampled set. The resulting variation is indicated on the plots as a band width around each curve. Datasets with a large number of samples, such as TissueNet, and BCSS, show a very small, hardly visible bandwidth on the curves below, reflecting the low sampling variability that comes with a large sample size. The PCam and CheXpert datasets have fewer samples, and consequently shows a visibly wider bandwidth resulting from this variation.

Table 1. Data Availability Statement containing dataset names, repositories, and accession details.

Dataset	Full Name	Accession / URL
MNIST	MNIST Database of Handwritten Digits	DOI: 10.24432/C53K8Q; `http://yann.lecun.com/exdb/mnist/`
BCSS	Breast Cancer Semantic Segmentation Dataset	DOI: 10.1093/bioinformatics/btz083; `https://github.com/PathologyDataScience/BCSS`
CheXpert	CheXpert Chest X-ray Dataset	`https://aimi.stanford.edu/datasets/chexpert-chest-x-rays`
PatchCamelyon	PatchCamelyon (PCam) Benchmark Dataset	DOI: 10.5281/zenodo.2546921
TissueNet	TissueNet Cell Segmentation Dataset	DOI: 10.1038/s41587-021-01094-0; `https://datasets.deepcell.org/`

PCam: tumor classification

The PatchCamelyon (PCam) benchmark dataset is a collection of histopathologic scans of lymph node sections [50] derived from the Camelyon 16 challenge [51] (see Table 1). PCam consists of 327,680 colored images of size 96x96 each. An arbitrarily chosen sample data is shown in Fig 4. The challenge is to identify the presence (positive) or absence (negative) of any histopathology in one image. The dataset creators of PCam emphasize that all the positive and negative instances in the training, validation, test sets are equally distributed, resulting in a class balance of 1:1. We used the pre-trained resnet34-pcam model, which is a popular, deep residual network from the TIA (Tissue Image Analysis) toolbox [52] with an overall f1-score of 0.889, as stated by the authors. We evaluated the uniformity of the p-values of the test samples and the performance of the FDR controlling methods. The results are shown in the first row of Fig 5. The Q-Q plots (Fig. 5-(i)-A) of the test EPVs (red dots) indicate that the test p-values are slightly biased, possibly indicating a slight overfitting and/or distribution shift between the training and test datasets. However, the Q-Q plots of the test EPVs adjusted with TNA- (green dots) and TNA+ (purple dots) show a more uniformly distributed p-values. The plots on Fig. 5-(i)-BCD indicate that the LTT and standard EPV methods yields mostly accurate FDR control with this dataset especially at critical α levels (0-0.1). The TNA methods (both TNA- and TNA+) yield p-values closer to uniform (Fig. 5-(i)-A), yet they result in conservative FDR control due to poor $\hat{\pi}_0$ estimation: the true $\pi_0 = 0.5$, while the Storey method estimated $\hat{\pi}_0 = 0.58$ for the standard EPVs, but $\hat{\pi}_0 = 0.65$ for both TNA- and TNA+.

Fig 4. Instances from PCam dataset.

In our opinion, the actual class proportions in real-life applications can be significantly different from 1:1, making the FDR control with methods like LTT inaccurate in practical applications. We manually resampled the test data so that the positive and negative class balance became 31%:69%, following the original class balance of the Camelyon 16 dataset. The evaluation plots are shown in the second row of Fig 5. The Q-Q plot (Fig. 5-(ii)-A) reveals that the EPVs (red dots) become slightly biased; however, our TNA methods manage to reduce this bias from the EPVs (green dots). The LTT method became liberal in FDR control, the BH protocol with standard EPVs resulted in conservative FDR control; whereas, the BH with the TNA EPVs remained accurate, especially at critical range of q-values (0-0.1) (Fig. 5-(ii)-C).

The bootstrap analysis shows moderately little variation for the PCam dataset, reflecting its relatively large sample size.

(i) Standard PCam dataset:
(ii) PCam with class distribution shift:
(iii) Standard CheXpert dataset:
(iv) CheXpert dataset with class distribution shift:
A B C D

Fig 5. Performance evaluation of FDR controlling methods. See the caption of Fig 3 for plot descriptions. The bandwidth around the black ground truth line indicating the variation from bootstrapping is indicated with a gray dotted.

Fig 6. Instances of CheXpert dataset.

CheXpert: chest x-ray dataset

The CheXpert dataset (see Table 1) contains a total of 224,316 radiographs of both frontal and lateral views of the thorax from ∼65K patients in order to facilitate the automated interpretation of chest X-rays for multi-label classification tasks (i.e. multiple labels can be predicted for each instances) with a total of 14 labels [53]. An arbitrarily chosen sample data is shown in Fig 6. We used the deep neural network model, called DenseNet121, from the `Torchxrayvision` package [54, 55], specifically developed for this CheXpert dataset. For the sake of simplicity, we turned the CheXpert dataset into a binary classification task by treating all instances of pleural effusion as positive and all the others as negative. We selected this label because this was the most prevalent disease in the dataset. The proportion of the positive class became 27 % and 23 % in the training and test datasets, respectively. The DenseNet121 model achieved accuracies of 89 % and 84 % on the training and test sets, respectively.

We evaluated the four FDR controlling methods and plotted the results on Fig 5-(iii). Interestingly, the most confident predictions of the DenseNet121 model in the test set are wrong. This can be seen by the flat region of the ground truth evaluation (black line) at low q-values around 0-0.05 (Fig, 5-(iii)-C). Fortunately, all four FDR controlling methods under investigations managed to capture this and remain close to the ground truth (Fig, 5-(iii)-B-D). The standard EPVs (red dots) tend to be off at the critical range (p-value < 0.1), but this does not affect the FDR controlling results in this benchmark, perhaps because of the confident incorrect predictions. Overall, we conclude that all four methods provide accurate FDR control. In our opinion, this indicates the absence of any database distribution shits. These plots also show that our TNA- and TNA+ algorithms do not corrupt the EPVs in the absence of data shift.

The actual frequency of pleural effusion may be different among all chest x-ray radiographs performed. It may vary over locations, age groups, in-patients and out-patients [56, 57]. In the United States, approximately 275 million conventional radiology procedures are performed annually [58], but only around 1.5 million of US citizens develop pleural effusion [57]. Pleural effusion develops in about 20-40% of in-patient with pneumonia [59]. Whatever the real abundance of the pleural effusion is among chest x-rays performed, it can be significantly different from the 27 % of the CheXpert dataset. Taking this into account, we re-sampled the test set in the CheXpert dataset to achieve a lower positive class ratio of 13%. The results of the re-evaluation are shown in Fig 5-(iv). The LTT and the TNA- EPVs are both slightly biased. The BH with the standard EVPs and with TNA+ EPVs achieve the most accurate FDR control over the entire range of α levels (0-1) in general. The LTT method is very

sensitive to class distribution shift demonstrated by this case, and it resulted in a liberally biased FDR control.

The CheXpert dataset has the smallest test set, and accordingly it shows the highest variability, both in the ground truth and in the estimated curves, as indicated by the bootstrap band widths on the figures.

TissueNet: cells segmentation

Fig 7. Instance of TissueNet dataset. Two-channel original images are being processed by the model, in turn producing masks & flows to correctly segment cells

TissueNet dataset contains 1.3 million microscopic images of cells obtained from six various platforms and nine organs including both histologically healthy and diseased tissues [60](see Table 1). Each images has size of 512x512 pixels, was manually segmented at the pixel level and paired whole-cell and nuclear annotations. An arbitrarily chosen sample data is shown in Fig 7. Our binary classification task was to predict whether a pixel is inside a cell (positive) or not (negative). The overall positive-negative ratio was 59%:41% in the training data and in contrast to 66%:34% in the test data. However, the class ratio varies over the images considerably. We used the Cellpose pre-trained deep neural network model [61] and fine-tuned it for five epochs. It achieved accuracy of 95% and 94% on training and test data, respectively. Our evaluation is reported in Fig 9-(i). The plots reveal that the BH protocol with standard EPVs becomes liberally biased (Fig. 9-(i)-B-C). The latter possibly originates from the data distribution shift or from overfitting. Both LTT and the BH with TNA EPV remain accurate.

The situation changes when we evaluate the segmentation problem with a single image. We arbitrarily chose one test image (ID: 007-008-009) and run the Cellpose model to classify each pixel. The positive and negative class balance was: 33%:67%. The Q-Q plot (Fig. 9-(ii)-A) of the test EPVs (red dots) indicates that the scores of negative samples from training and test datasets are different. The TNA EPVs on the Q-Q plot shows that the TNA methods have managed to adjust the p-values of the test samples (green and purple dots) closer to the uniform distribution. The plots on Fig. 9-(ii)-B-D indicate that LTT has turned out to be liberally biased, the BH method with standard EPVs strongly conservative. The BH with TNA- EPVs (green) and with TNA+ EPVs (purple) remain accurate in the critical region of q-values (0-0.4). Thus, the TNA methods have succeeded in correcting the p-values at the critical range at least.

BCSS: Breast cancer semantic segmentation dataset

Fig 8. An instance from BCSS dataset.

The Breast Cancer Semantic Segmentation (BCSS) dataset [62] contains around 20K image segmentation objects annotated through Digital Slide Archive, with a particular focus on tissue regions derived from breast cancer imagery obtained from The Cancer Genome Atlas (TCGA)(see Table 1). The image pixels are labeled by: 'Stroma' (class id: 0), 'Tumor' (class id: 1) and 'Others' (class id: 2). An arbitrarily chosen sample data is shown in Fig 8. The corresponding class ratios in the training and test sets are 56%:38%:6% and 33%:59%:8%, respectively. We used the deep learning model called `'fcn resnet50 unet-bcss'` from the TIA (Tissue Image Analysis) toolbox [52] to classify the pixels to one of the three classes. The model achieved an accuracy of 84 % and 82 % on the training and test datasets, respectively.

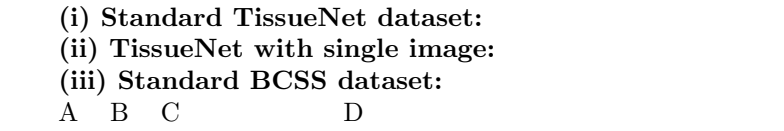


Fig 9. Performance evaluation of FDR controlling methods. See the caption of Fig 3 for plot descriptions. The bootstrap bands are narrow and hardly visible on these curves, due to the large sample size of the TissueNet and BCSS datasets, which results in little sampling variation.

Since this dataset involves a multi-class classification problem in which the FDR control is not straightforward due to the competition nature of the class prediction; thus, we slightly modified the FDR control procedure. We calculated the p-values, the $\hat{\pi_0}$ estimations with Storey method with respect to a single class (deemed as positive); and repeat it for each class via one-vs-rest fashion. Thus, for each pixels, we got three p-values, each corresponding to one of the three classes. Then we concatenated all of these p-values and calculated the q-values. We note the FDR control in this case allows the prediction of a single pixel into multiple classes. This is essentially a multi-label classification scenario.

The evaluation of the FDR controlling methods is shown in Fig 9-(iii). At first glance, the results suggest that LTT and TNA+ manage to control FDR well but this datasets is challenging for BH with standard or TNA- EPVs. The Q-Q plot of the EPVs exhibit deviations from uniformity but TNA EPVs are closer to being uniform (Fig. 9-(iii)-A). Overall, the BH control with standard EPVs is liberal at every alpha level (Fig. 9-(iii)-B); while the other methods are slightly liberal over the critical range of q-values (0-0.1) (Fig. 9-(iii)-C), and they become conservative beyond. BH protocol seems to be more accurate with standard rather than with TNA EPVs. The FDR estimation error with BH is significant (Fig. 9-(iii)-D).

Next, we further analyzed the results by each class separately. We plotted the training and test positive and negative discriminative score distributions class-wise provided by the deep learning model (Fig. 10 first row). The distribution plots show that the test scores are differently distributed from the training scores in all classes. We also plotted the distribution of the adjusted scores of the negative test samples, adjusted by our TNA methods (green and purple lines in Fig. 10) in order to see how well they are aligned to the training null distribution. In the case of the Stroma class, the mode of adjusted test scores (green and purple lines) move a bit closer to the one of the true training null distribution (X0, filled orange bars). However, their tails on the right hand side remain off. In the other two classes, the adjusted test scores align visibly better to the training null distribution (X0). The corresponding FDR curves are shown in the second row of Fig 10. Both the TNA- and the TNA+ methods improves the FDR estimation accuracy with all three classes. This is mainly due to the fact that the Storey method provides better $\hat{\pi_0}$ estimation using the TNA EPVs than using the standard EPVs. The LTT method is overly liberal with the Stroma class, and it is overly conservative with the Tumor class. We suspect that LTT appears to be accurate on the whole dataset simply by chance because the two types of bias cancel each other out in the Stroma and the Tumor classes. These two classes account for the the 94 % of the whole data.

In our opinion, the comparatively weaker performance of the TNA method on the BCSS dataset stems from subpopulation shift (proportions of different subclasses change from training to test sets). The shapes of the modes of the training and test null distributions differ more substantially than in the other benchmark datasets. In such cases, a simple affine adjustment of the test scores cannot adjust the test null mode close to the training null mode. The differing shapes of the modes across the three

classes can be seen in Fig 10. This divergence between the training and test null modes is most pronounced for the Tumor class.

Fig 10. Classwise score distributions and FDR control in the BCSS dataset.

Robustness to binning

Finally, we investigate the robustness of TNA to the hisotgam binning. We run TNA- and TNA+ over various numbers of histogram bins from 5 to 200, and compared the resulting numbers of trusted predictions to the ground truth. Results are shown on Fig 11. The results obtained with 30 bins or more are essentially the same; only the extremely coarse binnings (5–20 bins) deviate noticeably, because the density histograms become too crude to capture the shape of the null. TNA is therefore robust to the binning, and any sufficiently fine binning yields practically the same results.

Fig 11. Robustness to histogram binning. Top row: no shift; bottom row: class distribution shift.

Conclusion

In this study, we examined the effects of data shift on False Discovery Rate (FDR) control. In our opinion, the BH procedure with standard EPVs provides a very accurate FDR control in the absence of any data shift. Unfortunately, data shift is rather common in practice, and we demonstrated that the standard BH and the LTT methods are quite sensitive even to small data or class distribution shift. In this paper, we presented a simple, but robust heuristic method, called Test Null Adjustment (TNA), in order to mitigate the effect of the data shift on the FDR control. Our method operates in the space of prediction scores, thereby it is not affected by the curse-of-dimensionality problem, which often hinders methods that aim to mitigate data shift in the input feature space. The TNA method consists of two main steps. First, it approximates the null distribution among the scores of the test instances via certain statistics from the scores of the training instances. Second, it recalibrates the test scores so that the test and training null distributions would move closer to each other. TNA is fully data-driven, it does not rely on any theoretical assumptions or forms of the data. We have demonstrated in our experimental test with four biomedical datasets that TNA can successfully mitigate the bias caused by data shift; while, leaving the distributions intact in the absence of data shift. We note that, the handling of data shifts in practice is challenging, and unfortunately our method does not provide any theoretical guarantee that the FDR control with TNA EPVs becomes accurate, or conservatively or liberally biased without any additional assumptions; hence TNA is heuristic in its general form.

With this study, we also aim to emphasize and popularize the importance of accurate error control in any predictions in the biomedical domain. We have shown that the deviations in the simple Q-Q plots of the calculated vs. the theoretical p-values can be used to detect data shifts between the training and test data. Moreover, we also hope that the FDR control and plots with q-values will become a standard metric in performance evaluation for prediction systems in the biomedical domain.

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September 21, 2026

Manuscript: PONE-D-26-04885R1
"Toward reliable false discovery rate control in classification problems under distribution shift"

Dear Dr. Yamamoto:

We thank you and the two reviewers for the further careful evaluation of our revised manuscript.

Reviewer 2 recommends accepting the manuscript as it is. Reviewer 3 raised three methodological and statistical points, which we address in detail below. In brief:

1. *Asymmetric treatment of null and signal scores.* This comment led us to discover an implementation error, introduced during the previous revision through a miscommunication within our team (Step 5 of the TNA algorithm). We have restored the intended behavior, fixed the bug, and rerun all experiments to make sure that our results are reproducible and not error sneaked in the experimental results. The conclusions are unchanged, and the corrected figures are hardly distinguishable from the previous ones.
2. *Scope of FDR control.* We clarify, in the Introduction, that the multiple testing addressed in this work is over test instances (samples) and not a feature-selection problem, and we explain how the pixel-level classification tasks fit into this framing.
3. *Statistical rigor.* We carried out the requested bootstrap analysis, resampling both the training and test sets, and now show the resulting variability as bands around the curves in the figures. The variability is small overall, and decreases as expected with increasing dataset size.

In what follows, the editor's and reviewers' comments are shown in black type, interleaved with our responses (in blue) and, where appropriate, the modified or added text (in red).

Thank you very much for your consideration.

Best regards,

Prof. Attila Kertesz-Farkas
HSE University

Academic Editor – Journal Requirements

1. If the reviewer comments include a recommendation to cite specific previously published works, please review and evaluate these publications to determine whether they are relevant and should be cited. There is no requirement to cite these works unless the editor has indicated otherwise.

Neither Reviewer 2 nor Reviewer 3 recommended specific previously published works to cite, so no changes were necessary on this point.

Reviewer #2

The authors have addressed all the concerns I raised regarding the previous version. Therefore, I recommend accepting this manuscript as it is.

We thank the reviewer for the careful re-evaluation of our manuscript and for the positive recommendation.

Reviewer #3

We thank the reviewer for the careful and critical reading of our manuscript, and for raising these methodological and statistical concerns. We have split the reviewer's comments into three points and address each of them individually below.

1. What is the statistical rationale for assuming that the distributional shift only affects the null component and not the signal component (positive scores)? If a batch effect or a shift in scale occurs, it typically affects the entire distribution. The authors must provide a theoretical proof or a robust simulation-based argument showing why this asymmetric approach does not introduce bias into the detection of true positives.

We thank the reviewer for this comment. This comment definitaly leads to an interesting discussion; it touches on a few related aspects of TNA, so we address it below from several angles.

In our response preparation, we looked at Step 5 of the TNA algorithm and found that the previous revision contained an implementation inconsistency there: the corrective affine adjustment was applied only to test scores with a negative sign ($t_s < 0$), while non-negative scores were left unadjusted. This was not present in our original submission; it was introduced during the previous revision as a result of a miscommunication within our team, and it is an error. Restricting the adjustment to one side of zero can reorder test instances relative to one another (an adjusted negative-sign score can end up larger than an untouched positive-sign score), which then also distorts the ground-truth reference computed from the true labels against this reordered set of scores. We are grateful to the reviewer for a comment that led us to find and fix this.

We have corrected (reverted) Step 5 so that the affine adjustment is applied uniformly to every test score as in our original submission, regardless of sign: $\hat{t_s} = (t_s - \mu_{\hat{T_0}})/\sigma_{\hat{T_0}} \cdot \sigma_{X_0} + \mu_{X_0}$, for all $t_s \in T_s$. Since this is now a single, monotonically increasing affine map applied to the entire test score set, it preserves the relative order of all test instances – no reordering artifact can occur – and it recalibrates the whole test score distribution, not just its sub-threshold part, toward the location and scale of the training null. Therefore, TNA does not treat the shift as confined to the null component; the corrected Step 5 adjusts the entire score distribution using the same transform.

We re-ran all experiments reported in the manuscript with the corrected implementation. The qualitative conclusions are unchanged: the corrected figures are hardly indistinguishable from the previous ones, and all comparisons between methods reported in the manuscript remain valid. We have updated the manuscript accordingly (see Step 5 in the "TNA protocol" section).

A few of the broader statistical points raised by this comment remain relevant independent of this fix, which we address below.

(A) On statistical power under data shift: the theoretical guarantee of the Benjamini–Hochberg procedure (and of the underlying p-value machinery in general[? ?]) depends only on the p-values of the *truly null* instances being (super-)uniformly distributed; no analogous requirement is placed on the p-values of the truly non-null (positive, signal, alternative hypothesis) instances. However, we note that we do not have any guarantees on the statistical power. Depending on the shift, it can result in either decreased power (if the test non-null scores move closer to the training null X_0, yielding larger p-values for the true positives) or increased power (if the test non-null scores move further away from X_0, yielding smaller p-values for the true positives).

In the manuscript, in the TNA protocol, and in our experimental validation, we do not claim that the prediction accuracy remains the same (as in the training dataset); and we do not claim that we can fix wrong predictions for samples which were misclassified due to data shift. Our aim is to make the FDR control at the desired level less biased under data shift.

(B) On the distributional assumptions of TNA: we would like to point the reviewer to the "TNA protocol" section of the manuscript, which already states precisely what TNA- and TNA+ assume, we quote it below:

> "The first variation, called TNA- (TNA minus), relies on the assumption that the shapes of the training null (X_0) and test null (T_0) distributions are 'similar', but there is no assumption on the form of the non-null distributions. The second variation, called TNA+ (TNA plus), relies on that the shapes of the training non-null (X_1) and test non-null (T_1) distributions are 'similar', but there is no assumption on the analytical form of the null distributions. By 'similar' distributions we mean that they are close in a statistical sense, such as having a small Wasserstein distance or Jensen-Shannon divergence; intuitively, their histograms can be aligned without major reshaping. TNA's performance is robust to minor distributional shifts in practice, but it degrades gracefully as this distance increases. The proportion of the nulls may freely vary between the training and test data in the case of both TNA methods, there is no assumption on this."

That is: TNA places its similarity assumption on only one component (null for TNA-, non-null for TNA+), leaving the other component's shift unconstrained, and the manuscript states that TNA degrades as the similarity

decreases between training and test. The user can pick TNA- or TNA+ depending on whether the null or non-null component is expected to be more stable (respectively).

(C) We note that, we already have simulation-based evidence, in the manuscript, of TNA behaving well under a shift that plausibly affects the entire score distribution rather than the null alone: in Example 2 (MNIST with a 10% reduction of pixel intensity applied to every test image), the perturbation acts on all test images regardless of their true label, so both the null and non-null test scores are affected; TNA nonetheless recovers accurate FDR control (Fig 3(i)).

(D) On theoretical guarantees: we already discussed this with another reviewer in the first round; we summarize that response below for completeness (see our previous Response to Reviewers, Reviewer 2, point 1, for the full discussion):

> TNA is presented as a practical heuristic, not as a method with proven error bounds. We state in the manuscript that our method does not provide any theoretical guarantee that the FDR control with TNA EPVs becomes accurate, or conservatively or liberally biased, without additional assumptions; hence TNA is heuristic in its general form. To our knowledge, no existing method for FDR control under unconstrained, general distribution shift currently offers such guarantees either; we therefore frame TNA as a practically useful, model-agnostic alternative rather than as a theoretically certified procedure. More fundamentally, most FDR-controlling procedures (e.g., BH) only carry their theoretical guarantee under the assumption that the input p-values are valid, i.e., uniformly distributed under the null; in practice, this validity typically relies on an assumed parametric null (most commonly a standard normal distribution) that rarely holds exactly for real data, so FDR control breaks down not because the procedure itself is flawed, but because its premise is violated. TNA targets exactly this gap by estimating the null empirically from the training data instead of assuming a parametric form; the price of dropping that assumption is that we can only offer a heuristic argument rather than a formal guarantee, a trade-off already reflected in the manuscript's title, "toward reliable FDR control" rather than "guaranteed FDR control".

If the reviewer referred to something else then we kindly ask him/her to elaborate or point to the relevant paragraph/sentence of the manuscript.

2. Conceptual Ambiguity Regarding FDR Control: The manuscript exhibits a fundamental confusion between FDR control in the context of 'Feature Selection' (multiple testing across features) and 'Sample-level Inference' (calibrating significance for a single test). The authors fail to clarify whether the primary objective of TNA is to improve feature selection stability or to ensure the validity of p-values for individual samples. This conflation undermines the statistical interpretability of the results. The authors must explicitly state the scope of their FDR control and demonstrate how the TNA method accounts for the multiplicity of tests if applied to feature selection, or conversely, justify why a sample-level adjustment is sufficient for high-dimensional feature discovery.

We thank the reviewer for asking us to clarify this, and we would like to clarify: we do not perform, nor do we make any claim about, feature selection anywhere in this manuscript. TNA and the FDR control built on top of it operate entirely at the level of test *instances* (samples), not at the level of features. We agree that the manuscript should state this scope explicitly rather than leave it to be inferred.

We add the clarifying sentences below to the Introduction. We state this in terms of the general set-up (test instances and their scores), without yet referring to our method by name, since it has not been introduced at that point in the manuscript.

> We emphasize that the multiplicity addressed throughout this paper is across the m test instances $t_1, \ldots, t_m$, each associated with its own null hypothesis ("instance t_i belongs to the negative class"); it is not a feature-selection problem, in which the multiplicity would instead be across a set of candidate features or variables. Each score t_i is a single, already-computed scalar value produced by a machine learning classifier for one test instance, not a property of an input feature, and the classifier's input features play no further role once this score is obtained.

We would like to note one further nuance, which we think is worth making explicit rather than leaving implicit. In two of our datasets, TissueNet and BCSS, the classification task is performed at the pixel level: each pixel of an image is treated as one test instance, and the model predicts a class for every pixel individually. Since the pixels are themselves the raw input to the image, the multiple testing in these two datasets is, in a sense, simultaneously over test instances ("samples", in machine-learning terminology) and over input features (pixels, in a feature-selection-style terminology); the same underlying set of entities can be described either way. However,

pixel-wise image classification and semantic segmentation are conventionally discussed in the computer vision and machine-learning literature by treating each pixel as an individual classification instance (a sample), rather than by using feature-selection terminology, in which pixels would instead be framed as candidate features to be individually tested and selected. To avoid needlessly complicating the exposition, and to remain consistent with how this task is conventionally described in that literature, we discuss pixel classification throughout the manuscript as a machine-learning (sample-level) problem rather than as a feature-selection problem; we did not intend to imply anything about feature-selection multiplicity by this choice of terminology.

3. While the manuscript introduces TNA to manage FDR under distributional mismatch, the results currently lack sufficient statistical rigor. To address this, the authors should employ bootstrapping procedures on both the training and test sets to generate and report confidence intervals for all performance metrics.

We thank the reviewer for this suggestion and have carried out the requested bootstrap analysis. In each of 100 replicates, we jointly resampled, with replacement, both the training data (from which X_0 and X_1, and hence the TNA correction, are estimated) and the test data, and recomputed the entire pipeline – the EPVs, the TNA correction, and the BH/LTT-based curves – on each resampled pair. We report the resulting variability as a shaded band (bootstrap percentile interval) around each of the previously reported curves in Figs 5 and 9.

The outcome for the datasets with comparatively small test sets (PCam, CheXpert), the bootstrap bands are narrow but visible, reflecting genuine finite-sample uncertainty. For the datasets with much larger test sets (TissueNet, BCSS), the bands are so narrow that they are not visually distinguishable from the point-estimate curves themselves, since the variability of the underlying histogram-based estimates shrinks as the number of test (and training) instances grows. This confirms that the curves reported throughout the manuscript are not artifacts of a particular train/test split, and that their variability, where present, decreases exactly as expected with dataset size. We included the following text to the beginning of the results section:

> We qualified the stability of the methods with bootstrap analysis. For each dataset, we resampled both the training and the test sets 100 times with replacement, and recomputed the standard EPV, the TNA, and the LTT methods on each resampled set. The resulting variation is indicated on the plots as a band width around each curve. Datasets with a large number of samples, such as TissueNet, and BCSS, show a very small, hardly visible bandwidth on the curves below, reflecting the low sampling variability that comes with a large sample size. The PCam and CheXpert datasets have fewer samples, and consequently shows a visibly wider bandwidth resulting from this variation.

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